

John Benjamins Publishing Company



This is a contribution from *Multi-Dimensional Analysis, 25 years on. A tribute to Douglas Biber*.

Edited by Tony Berber Sardinha and Marcia Veirano Pinto.

© 2012. John Benjamins Publishing Company

This electronic file may not be altered in any way.

The author(s) of this article is/are permitted to use this PDF file to generate printed copies to be used by way of offprints, for their personal use only.

Permission is granted by the publishers to post this file on a closed server which is accessible to members (students and staff) only of the author's/s' institute, it is not permitted to post this PDF on the open internet.

For any other use of this material prior written permission should be obtained from the publishers or through the Copyright Clearance Center (for USA: www.copyright.com).

Please contact rights@benjamins.nl or consult our website: www.benjamins.com

Tables of Contents, abstracts and guidelines are available at www.benjamins.com

Dimensions of register variation in Brazilian Portuguese*

Tony Berber Sardinha, Carlos Kauffmann &
Cristina Mayer Acunzo
São Paulo Catholic University, Brazil

In this chapter, we present an MD analysis of Brazilian Portuguese, based on the Brazilian Register Variation Corpus, comprising 48 different spoken and written registers. The previous literature on text varieties in Brazilian Portuguese focuses on specific aspects of one or at most a few varieties at a time and, therefore, does not present a comprehensive picture of register use in Brazilian Portuguese. In this study, we attempt to fill this gap by employing the MD framework, which enabled us to account for a large number of different registers, based on a wide range of linguistic features. The analysis revealed six dimensions of variation, which are presented and discussed in this chapter.

1. Introduction

The MD framework has been used to analyze a growing number of world languages, including English (Biber 1988; Crossley & Louwerse 2007; de Mönnink, Brom, & Oostdijk 2003; Lee 1999), Korean (Kim & Biber 1994), Somali (Biber & Hared 1994), Nukulaelae Tuvaluan (Besnier 1988), Gaelic (Lamb 2008), and Spanish (Asención-Delaney, in this volume; Biber, Davies, Jones & Tracy-Ventura 2006; Biber & Tracy-Ventura 2007; Parodi 2007). In this chapter, we report on a language-wide MD analysis of Brazilian Portuguese. A Romance language with more than 200 million users around the world, Portuguese ranks sixth in terms of native speakers, of which approximately 90% are Brazilian.¹ Several distinctive differences exist among the varieties of Portuguese in terms of lexis (Bacelar do Nascimento, Mendes, Antunes, & Pereira 2014; Kilgariff, Jakubíček, Pomikalek,

* A version of the research reported here appeared in *Corpora*, 9 (2).

Berber Sardinha, & Whitelock 2014), syntax (Castilho 2009, pp. 883–885), and register use, which led us to restrict the scope of our study of this language to only one of its varieties: Brazilian Portuguese.

The research findings from MD studies provide a rich picture of the landscape of language use by the populations served by the different languages. In turn, the particular dimensions for each individual language capture the breadth of variation in these languages, revealing the principal parameters underlying the variation, which in many cases are obstructed from view by the profusion of texts in the environment. At the same time, MD research findings also enable us to compare the languages with respect to the dimensions, in some cases revealing how languages that one would assume were worlds apart do in fact have points of contact that were hitherto unknown. These cross-linguistic comparisons seem to point to the existence of so-called universals of register variation, which are dimensions that recur across many different languages (see Biber, this volume.) This study lends further support to the hypothesis of universal dimensions of register variation, as discussed below, while simultaneously revealing dimensions specific to Brazilian Portuguese, reflecting its unique register landscape.

Previous studies have examined the textual varieties of Brazilian Portuguese from a range of perspectives, including discourse analysis (Souza e Silva & Brait 2013), variation (Preti 2005), and corpus-based (Berber Sardinha 2005). However, most studies on the registers and genres of Brazilian Portuguese analyze a few varieties at a time, generally focusing on a small set of linguistic characteristics. The main goal of the present study is to fill this gap by looking at a wide range of registers, both written and spoken, based on a broad set of linguistic features, and in so doing identify the dimensions underlying register variations in Brazilian Portuguese. The only other MD analysis of Brazilian Portuguese to date is Kauffmann's (2005) analysis of newspaper registers, based on a carefully constructed corpus comprising 14 different registers, which identified two dimensions: Narrative versus Expository, and Argumentation versus Informational.

2. Methods

We followed the method set forth by Biber (Biber 1988; Friginal & Hardy, this volume), and revised the existing literature on structural characteristics of Portuguese (e.g. Azevedo 2005; Bechara 1999; Castilho 1989; Cunha 2001; Ilari 1991; Moura Neves 2000; Thomas 1969; Whitlam 2010), collected a register-diversified corpus, tagged it electronically for part-of-speech and lexical features, computed the frequencies of the individual features, processed these counts statistically, and interpreted the resulting factors qualitatively in terms of the communicative functions

performed by the co-occurring groups of features. As with all MD studies comprising a whole language, we encountered numerous challenges, the first and perhaps least obvious of which was corpus design. Here the researcher is faced with a series of decisions that will affect the outcome of the research. Our choice was to collect a corpus that was as diverse as possible in the number of different registers it comprised, which would allow us to represent the extent of the variation in Brazilian Portuguese as faithfully as possible. In so doing, we included extant print registers such as fiction literature, newspapers, and academic writing, as well as online communications like email, Twitter, and Facebook and everyday run-of-the-mill texts such as *bulas* (drug labels or drug information inserts), user manuals, and packaging labels. We wanted our corpus to be as realistic as possible from a language user's perspective, covering the daily language encounters of average Brazilian speakers. Another challenge was tagging and counting the individual linguistic features; to accomplish this, we employed PALAVRAS, a state-of-the-art parser (Bick 2014) that annotates more than 300 different features, including morphology, syntax, and semantics. For counting, we developed our own software,² to which we fed the tagged corpus and our detailed hand-crafted search patterns (based on individual tags and tag combinations) and whose output we carefully revised for accuracy. The extraction of correlated features (the factors) and their interpretation was also challenging, which depended on understanding the complex relationships among the linguistic features in and across the texts.

Our corpus, named the Brazilian Register Variation Corpus (CBVR; Corpus Brasileiro de Variação de Registro), was compiled specifically for this research project, but our intention is for it to be used for other projects as well (e.g. Berber Sardinha, São Bento Ferreira, & Teixeira 2014). It comprises 48 different registers (5,644,006 words total): 12 spoken (1,547,853 words, 27.5%) and 36 written (4,096,153 words, 72.5%). The composition of the corpus appears in Table 1.

The design of the corpus was guided by two main principles – namely, coverage and balance. The corpus includes as broad a representation of the registers in contemporary Brazilian Portuguese as possible, from the most prestigious (e.g. novels) to the most prosaic (e.g. product labels). The corpus is balanced in terms of the number of texts per register, with each register having exactly 20 texts. However, whenever individual texts were shorter than 400 running words, they were joined in the same file by other texts from the same register until the word count for the file reached at least 400. Therefore, the 20 files in each register might

2. Two tools are available to download for counting tags in texts parsed by PALAVRAS: Tag Count and CG3-IDE. They are both available on <http://corpuslg.org/gelc/gelc.php/2013/05/25/dimensions-of-variation-in-brazilian-portuguese>.

Table 1. Composition of the Brazilian Register Variation Corpus (CBVR)

Register*	Words	% of total words
1. Academic articles	92,148	1.6%
2. Agreements	44,562	.8%
3. Blogs	31,486	.6%
4. BUSINESS CONFERENCE CALLS	106,076	1.9%
5. Business letters	12,720	.2%
6. Campaign plans	29,724	.5%
7. Church liturgy	66,995	1.2%
8. Comics	25,937	.5%
9. CONGRESSIONAL DEBATES	641,080	11.4%
10. CONVERSATION	93,470	1.7%
11. Editorials	11,233	.2%
12. Emails – Personal	11,223	.2%
13. Encyclopedia entries	13,690	.2%
14. Essays (a)	21,403	.4%
15. Facebook	11,022	.2%
16. Game instructions (b)	16,260	.3%
17. General fiction	403,796	7.2%
18. Government bids (c)	127,239	2.3%
19. Horoscope	12,637	.2%
20. INTERVIEWS – SOCIOLINGUISTIC	152,788	2.7%
21. INTERVIEWS – PRESS	28,175	.5%
22. INTERVIEWS TV	263,821	4.7%
23. Jokes	9,310	.2%
24. Legislation	125,531	2.2%
25. Magazine celebrity	25,738	.5%
26. Magazine news	19,850	.4%
27. Medicine/drug labels (d)	16,061	.3%
28. Minutes	25,929	.5%
29. Newspaper reportage	11,467	.2%
30. Non-fiction books	55,028	1.0%
31. POLITICAL SPEECHES	44,591	.8%
32. Prep. school texts (e)	15,411	.3%
33. Product labels	9,183	.2%

(Continued)

Register*	Words	% of total words
34. RADIO BROADCASTS	91,335	1.6%
35. Recipes	9,591	.2%
36. Short stories	57,362	1.0%
37. SOAP OPERAS	93,627	1.7%
38. SONGS	11,990	.2%
39. TEXTBOOK DIALOGS (f)	9,447	.2%
40. Textbook texts (g)	12,732	.2%
41. Textbooks	1,234,790	21.9%
42. Theses	617,943	10.9%
43. TV NEWS	11,453	.2%
44. Twitter	11,027	.2%
45. User's/owner's manuals	301,650	5.3%
46. Websites	28,338	.5%
47. Written exams	33,937	.6%
48. Youth fiction	543,200	9.6%
Total	5,644,006	100.0%

*Registers in UPPER CASE are spoken.

(a) Short narrative pieces, generally printed in newspapers (*crônicas*)

(b) Instruction leaflets that accompany board games

(c) Government contract announcements

(d) Information leaflets enclosed in medicine packages

(e) Texts from university entrance exam practice manuals (*apostilas de cursinho*)

(f), (g) Texts from a Portuguese as a Foreign Language course book.

actually comprise more than 20 source texts. This was a precaution against basing the word counts on samples that were too short to represent the range of linguistic characteristics actually occurring in the register.

Table 2 provides a comparison among the corpora used in previous MD studies. As can be seen, the CBVR is the largest among the studies surveyed in terms of the number of different registers, the second in the total number of texts, and third in terms of word count.

The literature review on the linguistic features to be included in the analysis revealed a large number of potential candidates. We drew a list of the possible features and evaluated each one based on its relevance to register variation as well as whether it lent itself to automatic identification.

The corpus was tagged with PALAVRAS, as mentioned, and the linguistic characteristics tagged comprehended the vast majority of the features we had identified in the literature review. The parser was run on the corpus, and several

Table 2. Comparison of Select Corpora Used in Previous MD Studies

Language	Registers (written/spoken)	Texts	Texts per register	Word count
1. English	23 (17/6)	481	6 a 80	960,000
2. English	22 (4/18)	N/A	N/A	6,287,734
3. Tuvaluan	7 (2/5)	222	12 a 70	152,771
4. Korean	22 (12/10)	150	5 a 10	135,500
5. Somali	33 (23/10)	604	3 a 49	600,000
6. Spanish	19 (8/11)	4049	16 a 791	20,301,847
7. Spanish	3 (2/1)	90	4 a 74	1,466,744
8. Brazilian Portuguese	48 (12/36)	960	20	5,644,006

Key: 1. Biber (1988); 2. Crossley & Louwerse (2007); 3. Besnier (1988); 4. Kim & Biber (1994); 5. Biber & Hared (1994); 6. Biber & Tracy-Ventura (2007); 7. Parodi (2007); 8. the present study.

samples were checked by hand to ascertain the quality of the annotation; whenever necessary, the annotation was amended. Features not reliably tagged were removed from the pool. The accuracy rate for the parser was estimated to be 95% for syntax, 98.6% for part of speech, and 99% for lemmatization (Bick 2014). The final inventory comprises 190 features (see Table A1 in the Appendix), of which 93 remained after the final factor extraction (see Table A2 in the Appendix.)

Once the corpus was fully tagged and the verification process completed, the features were counted using a program specially designed for this investigation. The program took the tagged output and applied search algorithms to it in order to identify features of interest. Three versions of the count program were produced: a Unix script, a Java and OSX version (named PalavrasCounter), and a self-contained executable script (called CG3-IDE; Didriksen, Berber Sardinha & Bick 2013). The first script is available upon request; the other two versions are freely available from the Corpus Linguistics Research Group (GELC) website at corpuslg.org/gelc. Following Biber’s (1988) recommendation, the feature counts were normed per 1,000 words to enable comparisons among texts of different sizes. Normed frequencies are calculated as follows:

$$(\text{observed frequency of feature in the text} / \text{total words in the text}) \times 1,000$$

For instance, editorial #13 has 54 adjectives out of a total word count of 463; hence, its normed count for adjectives is:

$$54 / 463 \times 1000 = 116.63$$

In contrast, textbook #10 has more than ten times as many adjectives (4,968), but it is also much longer (65,069 words); hence, its normed adjective count is:

$$4968 / 65069 \times 1000 = 72.20$$

In other words, editorial #13 has about 50% more adjectives per thousand words than textbook #10, even though the observed frequencies might suggest otherwise.

An initial factor analysis was then run on the normed data, using principal axis factoring as the extraction method on SPSS 20 for Mac. The data were screened through the Kaiser-Meyer-Olkin Measure of Sampling Adequacy, which yielded a score of .875, indicating that correlation patterns are noticeable in the data. In addition, Bartlett's Test of Sphericity was significant (chi-square = 78008.8, $df = 10731$, $p = .000$), meaning that the correlation matrix is well-formed. Both screening tests suggest that the data are suitable for factor analysis. The factoring then proceeded as follows. Features with communalities lower than .2 were dropped (following Biber 1995, p. 138 – a more conservative value than the .15 suggested in Biber 2006, p. 183), and a further factor extraction was conducted. A scree plot was generated and, upon inspection, it suggested the existence of six factors in the data (see Figure 1).

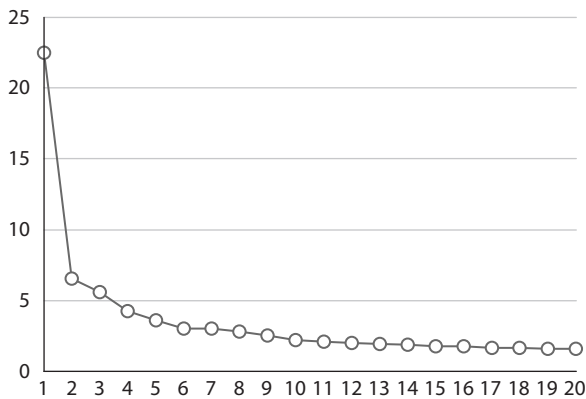


Figure 1. Scree plot

These six factors account for 31% of the variation in the corpus (see Table A3 in the Appendix for the amount of variance captured by each factor.) A rotated factor analysis was subsequently performed, with Promax rotation, which allows for some correlation among the factors. The factor inter-correlations (shown in Table A4 in the Appendix) are weak, ranging from $-.41$ to $.48$. These values fall within the normal range for MD studies: In English, they varied from $-.49$ to $.30$ (Biber 1988, p. 84), in Spanish, from $-.36$ to $.44$ (Biber et al. 2006, p. 37), and in Somali, from $-.54$ to $.51$ (Biber 1995, p. 129). With a rotation, 'each linguistic feature tends to load on only one factor, and each factor is characterized by those relatively few features that are most representative of the underlying construct'

(Biber 1988, p. 102). A cut-off was applied to the weights, whereby only those features with loadings equal to or greater than + or – .3 were used (after Biber 2006, p. 14). As is customary in MD analyses, several features loaded on more than one factor after the rotation; these were considered during the interpretation of the factors. However, a feature was only entered once in the computation of the dimension score – namely, for the factor on which it had the greatest absolute weight. In Tables 3 through 8, features that have larger weights on a different factor are enclosed in parentheses. The rotated factor pattern, with the loadings for individual features, is shown in the appendix (Table 4).

Each factor was then interpreted functionally, based on the communicative characteristics of the features loading on it (see Biber 1995, pp. 136–138). The interpretation of factors leads to the definition of the dimensions and is therefore a crucial step in an MD analysis. It is a complex process that requires consideration of the functions that features perform on different levels – namely, on its own, as a group together with other features, and across separate factors, again both individually and as a set (as features can occur on more than one factor). Regarding the need to consider all of these interactions among the features, Biber (1995, p. 135) remarked:

In many cases, features can have somewhat differing functions in different kinds of text; further, features can have functions defined at different levels of generality. The MD approach assumes the existence of shared functions underlying groupings of co-occurring features, so that at least some of the functions associated with a given feature will be shared with the other features defining a dimension.

In addition to considering the functions of linguistic features in the interpretation of the factors, the analysts should also ‘consider the similarities and differences among registers with respect to the set of co-occurring linguistic features.’ (Biber et al. 2006, p. 14). This requires the computation of dimension (or factor) scores for each text, as well as the calculation of the mean dimension score for each register. These means are plotted on charts showing the placement of a register on the factor relative to the other registers, thereby indicating how marked or unmarked it is with respect to the communicative construct underlying the dimension. The analysts should then use their knowledge of the context in which the registers are used to aid in the interpretation of the factor in terms of a dimension.

Before computing dimension scores, the normed values have to be standardized. Standardization is required to control for the different frequencies of individual features, so that both high and low frequency features have a similar status in the computation of the dimension scores. Without standardization, the presence of highly frequent characteristics (e.g. nouns and verbs) would define the dimension scores, rendering their co-occurrence with less frequent, yet important

characteristics (e.g. different clause types) irrelevant. Standardized, or Z-scores, were calculated as follows: For each feature, its normed frequency was subtracted from its mean normed frequency; the result was then divided by the standard deviation of the feature. Let's use the previous example of editorial #13, which has a normed frequency of adjectives equal to 116.63; adjectives have a mean normed frequency in the corpus of 60.5 and a standard deviation of 22.19 (both given in Table A2 in the Appendix), meaning the Z-score for adjectives in editorial #13 is:

$$(116.63 - 60.5) / 22.19 = 2.53$$

The Z-score indicates that the frequency of adjectives in editorial #13 is 2.53 standard deviations above the mean:

$$(2.53 \times 22.19) + 60.5 = 116.63$$

Z-scores were calculated in SPSS for Mac 20 using the Descriptives command. It is important to highlight that the standardized scores should not be used for the extraction of factors, but rather only in the calculation of factor scores. For factor extraction, the normed counts must be used.

Accordingly, dimension scores were computed by summing up the standardized scores of the features loading on each pole of a dimension and then subtracting the sum of the negative pole features from the positive ones (if a dimension has a single pole, then the subtraction is not necessary.) The following illustrates the computation of Dimension 5 scores:

$$(Zvbsubfut + Zcjou + Zvbfutpres + Zmddever + Zmdpoder + Zcjcond + Zadvlikl + Zcjcoorphr) - (Znplac + Zvbsubpast + Zadjaffi + Zvbimpf + Zvbpast)$$

Thus, sum up the Z-scores on the positive pole of factor 5 (shown in Table 7), then sum up the Z-scores on the negative pole, then subtract the former from the latter. The features in parentheses in Table 7 were not included in the computation of Dimension 5 scores because they had higher loadings on a different factor and, therefore, were entered in the computation of the score for that other factor.

3. Interpretation of the factors

Factor 1 (Table 3) comprises the greatest number of features, with 49 linguistic characteristics total (35 positive, 14 negative), accounting for the greatest variation (15.3%). The positive features are mostly verbs (10 features), adverbs (7), and pronouns (7). First-person verb forms and first-person pronouns (either in subject or object positions) both place a focus on the addressor, whereas second-person pronouns indicate an addressee focus. Both *QU* and yes–no questions generally

reflect turn initiation between addressor and addressee and are used ‘primarily in interactive discourse’ (Biber 1988, p. 106). Mental and private verbs are used to express one’s inner thoughts and feelings, thereby contributing to the ‘addressee focus,’ while communication verbs shift that focus to the other participants in the discourse. Possessives can also contribute to the personal nature of the dimension, as they signal how entities relate to individuals in terms of ‘ownership.’ Action verbs, on the other hand, highlight the events taking place in a more ‘concrete’ way than the mental and private verbs, which can perform a more ‘subjective’ role; however, these are frequently voiced in first person, thereby reinforcing the idea of this interplay between addressor and addressee. Progressive forms preceded by an infinitive (e.g. *vou estar fazendo*, ‘I will be doing’) mark an informal vernacular, sometimes uneducated future form. A related kind of informal future is the *ir* future (*vou fazer*, ‘I’m going to do’), which is typically associated with spontaneous discourse. Negation adverbs (both simple *não* and other related forms such as *nunca* and *jamais*) convey denials and rejections, which can signal how interlocutors negate propositions. Intensity adverbs, amplifiers, and emphatics all ‘mark heightened feeling’ (Biber 1988, p. 106), which again contributes to shaping the discourse as person oriented. Place, manner, and time adverbs are all ‘other oriented features,’ ‘reflecting the description of other people in particular places and times’ (Biber et al. 2006, p. 12), thereby fitting this understanding of the dimension as combining both a ‘self-’ and an ‘other-’oriented perspective.

Table 3. Dimension 1: Oral versus literate discourse

Feature [label]	Loading
Pronouns: Second person, in object position [prn2obl]	.910
Verbs: First person [vb1]	.826
Verbs: Mental [vbment]	.827
Verbs: <i>Ir</i> future [vbfutir]	.821
Pronouns: Second person singular, in subject position [prn2sngsubj]	.763
Verbs: Private [vbpriv]	.702
Verbs: Action [vbact]	.689
Adverbs: <i>Não</i> (no) [advnao]	.674
Pronouns: First person singular, in subject position [prn1sngsubj]	.633
Adverbs: Time [advtime]	.629
Pronouns: First person, object position [prn1obl]	.622
Pronouns: Quantifier [prnqtf]	.616
Adjectives: Evaluative [adjeval]	.612
Pronouns: Possessive [prnpos]	.605

(Continued)

Feature [label]	Loading
Adverbs: Intensity [advints]	.552
QU questions [qsqu]	.539
Adverbs: Amplifier [advampl]	.522
Adverbs: Emphatic [advemph]	.476
Que clause controlled by verb in indicative mood [vbqueindic]	.467
(Adverbs: Place [advpl])	.460)
Adjectives: Predicative position [adjpred]	.456
(Yes or no question [qsyn])	.421)
Verbs: Infinitive [vbinf]	.407
Adverbs: Manner [advmanner]	.373
Verbs: Communication [vbcomm]	.356
(Discourse marker [discmrkr])	.350)
Verbs: Gerund form, all [vbgerall]	.345
(Subject omission [subjdrop])	.343)
Modals: <i>Precisar</i> (need to) [mdprecisar]	.338
Que clause controlled by adverb [clqueeadv]	.337
Verbs: Progressive preceded by infinitive [vbproginf]	.330
(Verbs: Future subjunctive mood [vbsubfut])	.323)
Adverbs: Negative, except <i>não</i> [advneg]	.320
(Subordinating (conditional) clause [cjcond])	.313)
Pronouns: Nominal in subject position [prnnomsubj]	.308
Reduced progressive clause [vbprogphr]	-.309
Pronouns: Relative <i>qual</i> or <i>cujo</i> [prnqualcujo]	-.315
(Adjectives: Affiliative [adjaffi])	-.336)
Agentless passives [clpassless]	-.377
Adjectives: Relational [adjrela]	-.422
Past participle [vbpastprt]	-.497
Nominalization in subject position [nominlzsubj]	-.505
Adjectives: Topical [adjtopi]	-.511
Nouns: Abstract [nabst]	-.521
Average word length [wl]	-.529
Adjectives: Attributive position [adjattr]	-.589
Nouns: Compound [ncomp]	-.651
Articles: Definite [artdef]	-.739
Prepositions: All [prpall]	-.776

The negative pole of factor 1 includes mostly nominal features, such as nouns, adjectives, articles, and prepositions, which generally help present information in a condensed manner in the text. Nouns are the ‘primary bearers of referential meaning in a text’ (Biber 1988, p. 104), and a concentration of them increases informational density. In nominal groups, prepositions connect the individual elements and help integrate information in compact units. Past participials are used in agentless passive voice constructions, which contribute to raising the level of abstractness in the discourse by eliding the agent. Reduced progressive clauses also work in a similar fashion: by removing agency and simultaneously raising the condensation of information by latching an extra load of information onto the main clause. Abstract nouns and nominalization boost the abstract nature of the information. Adjectives qualify the information being presented and, in the attributive position, they further increase the amount of information in the nominal group. The relative pronouns *qual* and *cujo* are both markers of learned discourse, which is typically characterized by high levels of abstraction and information content. Long words mark precise lexical choices, such as those instantiated by abstract nouns, nominalizations, participles, and gerund forms. Based on the functional associations of the features on the register distinctions, we posit the interpretive label ‘Oral versus Literate Discourse’ for Dimension 1, with orality representing ‘on-line production, and the literate pole [...] careful, usually edited, written production’ (Biber 1995, p. 242). The mean dimension scores are presented in Figure 2.

Although the positive end of Dimension 1 comprises both written and spoken varieties, all but two of the spoken registers are featured there. Most of the highest-scoring registers are addressee-oriented, dialog-based texts, such as face-to-face conversation, soap opera scripts, and interviews, but among these are also ‘simulated conversation’ registers such as songs and Facebook messages. The sample in Example (1) illustrates the dense use of involved features, such as first-person verb forms, first- and second-person pronouns, mental and action verbs, and *ir* future forms in songs.

- (1) *Eu já fiz de tudo pra te convencer*
Mandei rosas vermelhas
Lindas pra você
Falei de amor
Fiz uma canção
A Lua se foi, nem vi o sol chegar
Acreditei que o tempo não ia passar

Translation:

I’ve already done everything to convince you
 Sent you red roses
 Beautiful ones for you
 Spoke about love

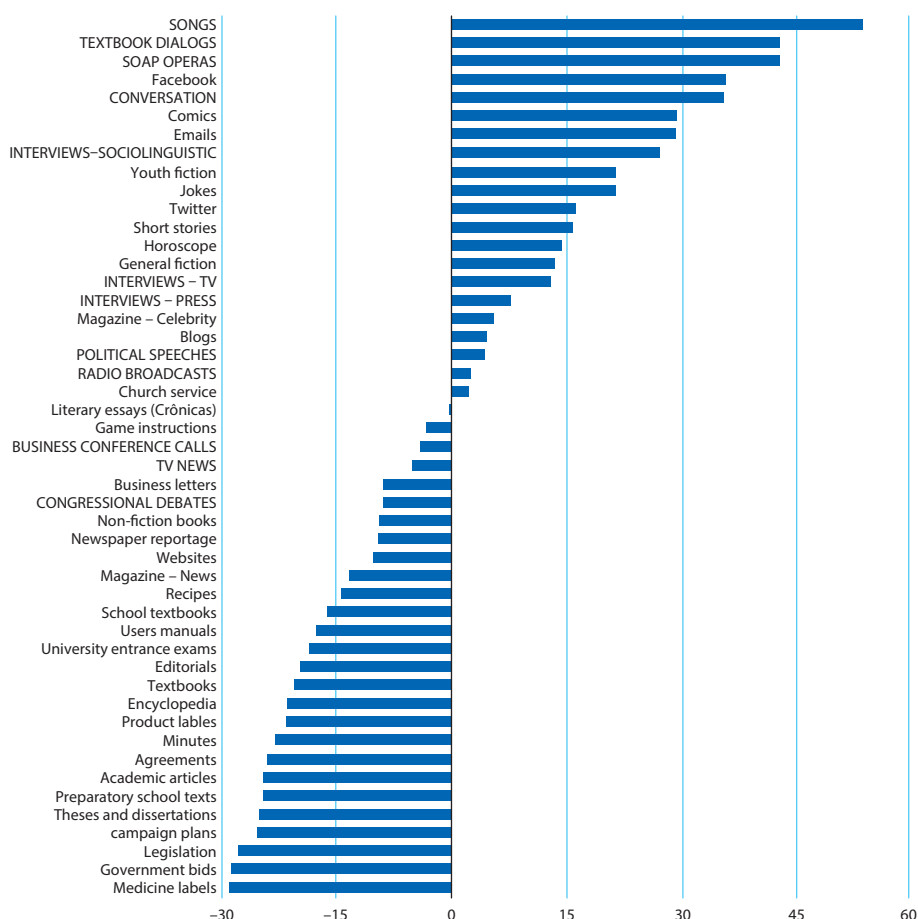


Figure 2. Dimension 1: Oral versus literate discourse

Wrote a song

The moon is gone, I haven't seen the sun arrive

On the negative pole are mostly written registers, loaded with informational content conveyed through prepositions, abstract nouns and noun compounds, nominalizations, adjectives in attributive position, and past participles, among other features. Example (2) is a sample from drug labels that illustrate the use of these features.

- (2) *Os comprimidos apresentam dois sulcos para facilitar a composição de doses. Indicações: [medicine name] é um glicocorticóide de uso oral, indicado para tratamento de diversas patologias, tais como: Reumatologia: artrite reumatóide, lupus eritematoso sistêmico (...)*

Translation:

The pills have two grooves that facilitate dosage adjustment. Prescription:
[medicine name] is a glucocorticoid that is to be taken by mouth,
prescribed for the treatment of numerous pathologies, such as:
Rheumatology: rheumatoid arthritis, systemic lupus erythematosus (...)

Factor 2 (Table 4) comprises only positive features. Most of the 19 linguistic characteristics are non-finite clauses, such as *que* (that/which) or infinitive clauses controlled by specific kinds of nouns or adjectives or by prepositions. In addition, the factor includes relative/adjectival clauses introduced by *que* relative pronouns. Clauses controlled by stance nouns or adjectives or by a particular class of adjectives (expressing ease or difficulty) are regularly employed to ascertain a point of view or to frame the information in a particular way. Relative clauses are ‘devices for the explicit, elaborated identification of referents in a text’ (Biber 1988, p. 110), which can then be commented on in the text in a manner that suits one’s framing. Demonstrative pronouns can also be used to single out particular referents, whether concrete or abstract. Comparative adverbs are commonly used as devices for evaluating entities or propositions and, as such, are valuable elements for

Table 4. Dimension 2: Argumentation

Feature [label]	Loading
<i>Que</i> clause controlled by noun [nounque]	.593
Pronouns: Relative <i>que</i> [prnque]	.529
Adverbs: Comparative [advcomp]	.473
Nouns: Cognition [ncogn]	.451
<i>Que</i> or infinitive clause controlled by noun (stance) [nqueinfcl]	.447
Infinitive clause controlled by adjective [clinfadj]	.447
<i>Que</i> clause controlled by preposition [clqueeprp]	.426
Pronouns: Demonstrative [prndem]	.406
Infinitive clause controlled by preposition [clinfprp]	.395
<i>Que</i> clause controlled by adjective (stance) [adjque]	.378
(Adjectives: Predicative position [adjpred]	.353)
(Pronouns: Quantifier [prnqtf]	.352)
(Pronouns: Third person, object position [prn3obl]	.343)
(Modals: <i>Poder</i> [mdpoder]	.336)
Infinitive clause controlled by ease or difficulty adjective [clinfadjease]	.334
Adverbs: Hedge [advhedg]	.331
Articles: Indefinite [artindef]	.325
Verbs: Future preterit tense [vbfutpret]	.311
Conjunctions: Coordinating (adversative) [cjadv]	.310

expressing personal attitude. Similarly, quantifier pronouns enable comparisons and thereby function as rhetorical devices as well. Coordinating adversative conjunctions introduce clauses that also mark comparisons and contrasts. Cognition nouns are rhetorical devices that convey abstract notions; they enable speakers to encapsulate complex information into a single word, which in turn affords greater control over how the information will be handled to achieve one's intentions in the discourse. Hedges express fuzziness or vagueness and, as such, they can also function as rhetorical devices. Indefinite articles indicate unspecified entities, thereby denoting a degree of abstractness. The modal *poder* expresses a variety of meanings, which include (but are not limited to) ability (*posso levantar isso*, 'I can lift that'), likelihood (*pode chover*, 'it might rain'), and permission (*Posso sair?* 'May I leave?'), all of which are useful for shaping propositions. Third-person pronouns in the object position mark a form of 'other direct discourse' detached from the immediate interlocutors. Adjectives in the predicative position are used to qualify particular entities. Future preterit forms are regularly used to formulate hypothetical statements. Put together, these features seem to mark the argumentative use of language; therefore, the proposed interpretive label for Dimension 2 is argumentation.

Figure 3 shows the register differences associated with Dimension 2. High scoring registers include not only standard argumentative debate-based registers such as political speeches, interviews (both print and TV), editorials, and congressional sessions, but also horoscopes, which are not commonly regarded as argumentative. Low-scoring registers, on the other hand, are defined by an absence of such features and, hence, of argumentation.

The sample in Example (3) shows the use of features associated with argumentation in a horoscope, such as different types of *que* clauses, comparative adverbs, demonstrative pronouns (or demonstrative determiners, in English grammar), relative clauses, adversative conjunctions, the modal *poder*, and cognition nouns.

- (3) *Siga a trilha que seus sentimentos propuserem hoje. Permita-se a surpresa, deixe de lado todos os planejamentos que tiver feito para este dia e observe a direção que seus sentimentos mais íntimos estabelecerem. Conversas sérias podem e devem ser desenvolvidas, mas antes de tudo você deve tirar de cima dessas o ar de gravidade, pois sem isso só encontrará resistência (...)*

Translation:

Follow the path that your feelings have laid out today. Allow yourself to be surprised, leave to one side all the planning that you have done for this day and note the direction that your most intimate feelings established. Serious conversations can and should be carried out, but before anything else you should take that air of gravity off them, because without that you will only find resistance (...)

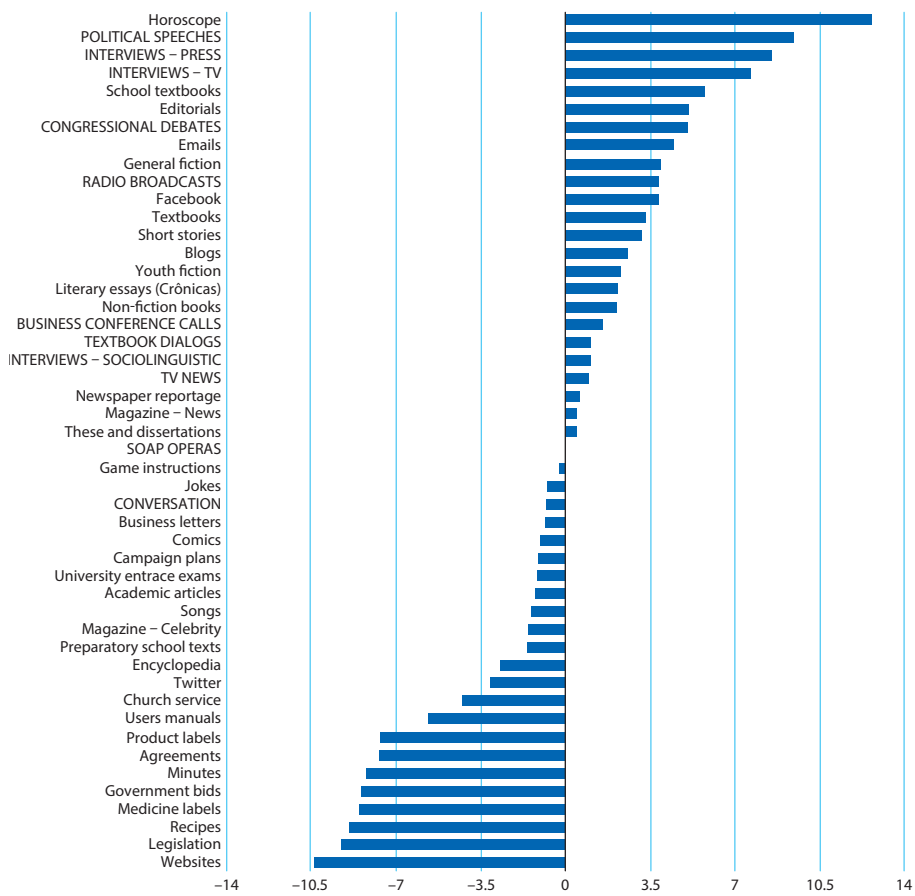


Figure 3. Dimension 2: Argumentation

Factor 3 (Table 5) incorporates 16 features on two poles, most of which have positive loadings (13). Tag questions and yes–no questions are both addressee-oriented features. Tag questions are interactive devices that can invite feedback from an interlocutor without a direct question. Yes–no questions, on the other hand, explicitly signal a change of turn, directly requesting the participation of the addressee. Third-person personal pronouns in both the subject and object positions are ‘other-oriented’ features that refer to participants outside the immediate context of the interaction. Conclusive conjunctions mark a specific logical relationship between clauses, indicating an inferential or deductive form of reasoning. Place adverbs typically mark a deictic reference so that the discourse is grounded in the immediate context. *Ter que* and *ter de* are necessity modals, expressing the speaker’s or writer’s stance in terms of an obligation. Demonstrative pronouns

signal a reference that is in the vicinity of the addressor. *Que* clauses controlled by a verb in the indicative mood are regularly reporting clauses or complements to a copula, such as *parecer* (to seem/appear). First-person personal pronouns refer directly to the speaker or writer. This set of features seems to point to a person-oriented, involved, interactive discourse that in many ways resembles the positive pole of both English Dimension 1 (Biber 1988) and Spanish Dimension 3 (Biber et al. 2006; Biber & Tracy-Ventura 2007).

Table 5. Dimension 3: Involved versus informational production

Feature [label]	Loading
Tag questions [qsttag]	.795
Contractions [contrac]	.714
Discourse marker [discmrkr]	.671
Questions: Yes or No question [qsyn]	.547
Pronouns: Third person singular, in subject position [prn3sngsubj]	.498
Pronouns: Third person plural, in subject position [prn3plusubj]	.481
Conjunctions: Coordinating (conclusive) [cjcnc]	.475
Adverbs: Place [advpl]	.462
Modals: <i>Ter que/ter de</i> (have to, ought to) [mdter]	.315
(Pronouns: Demonstrative [prndem])	.362)
(<i>Que</i> clause controlled by verb in indicative mood [vbqueindic])	.316)
(Pronouns: First person singular, in subject position [prn1sngsubj])	.314)
Type-token ratio [ttr]	–.346
(Adjectives: Attributive position [adjattr])	–.357)
(Pronouns: Possessive [prnpos])	–.431)

The negative pole of factor 3 has three features, only one of which has a higher loading on this factor, viz. type-token ratio, which measures vocabulary diversity. As such, texts with high ratios have a wide range of lexical choices. Adjectives in the attributive position are ‘used to further elaborate nominal information’ (Biber 1988, p. 105), reflecting a higher density of information. These features enable the conveyance of concentrated information. We therefore propose the interpretive label ‘Involved versus Informational Production’ to reflect the shared communicative functions of Dimension 3.

The mean registers scores for Dimension 3 appear in Figure 4. Sociolinguistic interviews and conversations are the most marked, followed by soap operas and TV interviews, thereby supporting the interpretation of the positive pole as reflecting involvement.

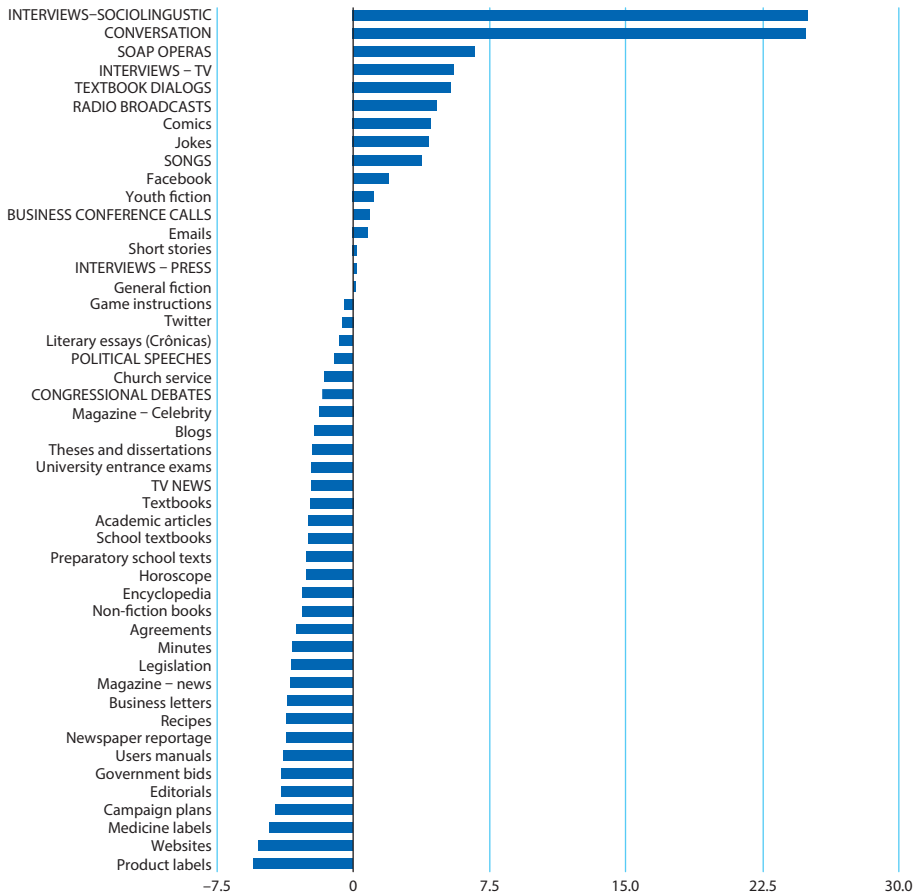


Figure 4. Dimension 3: Involved versus informational production

Example (4) is an excerpt of a sociolinguistic interview, in which contractions, question tags, place adverbials, and discourse markers are used.

- (4) A: *Então, o que que chocou na morte de Ayrton Senna, né?*
 B: *É, de repente, corrida, corrida pra mim seria o Ayrton Senna, né? Então, ligava-se a televisão pra assistir não a corrida propriamente dita, mas, de repente, pra ver alguém da gente, né? Brasileiro empunhando a bandeira lá.*

Translation:

A: So, that was shocking, Ayrton Senna's death, wasn't it?

B: Yeah, I mean, a race, a race to me would have to have Ayrton Senna, you know what I mean? So, you would turn on the TV not to watch the actual race, but, really, to see someone like us, you know? Brazilians hoisting the flag over there.

Example (5) presents an excerpt from product labels, from a text with a type-token ratio of .6, above the mean TTR of .53 for the corpus as a whole. Adjectives in the attributive position are underlined.

- (5) [Product name] é um soro concentrado anti-fadiga, com extratos de brócolis, soja e outros ingredientes naturais. Este soro com 97.5% de ingredientes naturais ajuda a potencializar o funcionamento da mitocôndria e produção de energia celular, reduzindo a fadiga da pele antes que vire sinal de envelhecimento. [Product name] 30ml é o cuidado altamente concentrado que proporciona excepcionais resultados antienvelhecimento.

Translation:

[Product name] is a concentrated anti-fatigue serum with extracts of broccoli, soy and other natural ingredients. This 97.5% natural ingredient serum helps enhance the functioning of the mitochondria and encourage cellular energy production, reducing skin fatigue before signs of aging set in. [Product name] 30ml is highly concentrated care that provides exceptional anti-aging results.

Factor 4 (Table 6) includes seven features on a single pole. Six variables load primarily on it, like the present subjunctive and the imperative moods, both of which commonly occur in pro-drop structures to express directives. Concrete nouns refer to ‘inanimate objects that can be touched’ (Biber 2006, p. 248), whereas facilitation or causation verbs ‘indicate that some person or inanimate entity brings about a new state of affairs’ (Biber 2006, p. 247). Clausal coordination links clauses but retains their independent equal status. Adverbs of manner convey ‘information about how an action is performed’ (Biber, Johansson, Leech, Conrad, & Finegan 1999, p. 553). The co-occurrence of these characteristics is frequently used to give instructions on how to perform tasks, many of which are hands on and achieve particular outcomes, usually of a practical nature; hence, we propose the interpretive label ‘directive discourse’ for Dimension 4.

Table 6. Dimension 4: Directive discourse

Feature [label]	Loading
Verbs: Present subjunctive mood [vbsubpres]	.821
Verbs: Imperative mood [vbimp]	.774
Nouns: Concrete [nconc]	.565
Subject omission [subjdrop]	.545
Verbs: Facilitation [vbfacil]	.485
Conjunctions: Coordinating (clausal) [cjcoorcls]	.465
(Adverbs: Manner [advmanner])	.345)

The register differences based on Dimension 4 are shown in Figure 5; the most marked register is recipes, which by their very nature require directives to explain how to prepare drinks and dishes. Owner's manuals and game instructions also provide directions for operating tools and appliances as well as for playing games. The negative end of Dimension 4 is characterized by registers with middle-range scores; for reasons of space and because they lack directive features, they will not be illustrated.

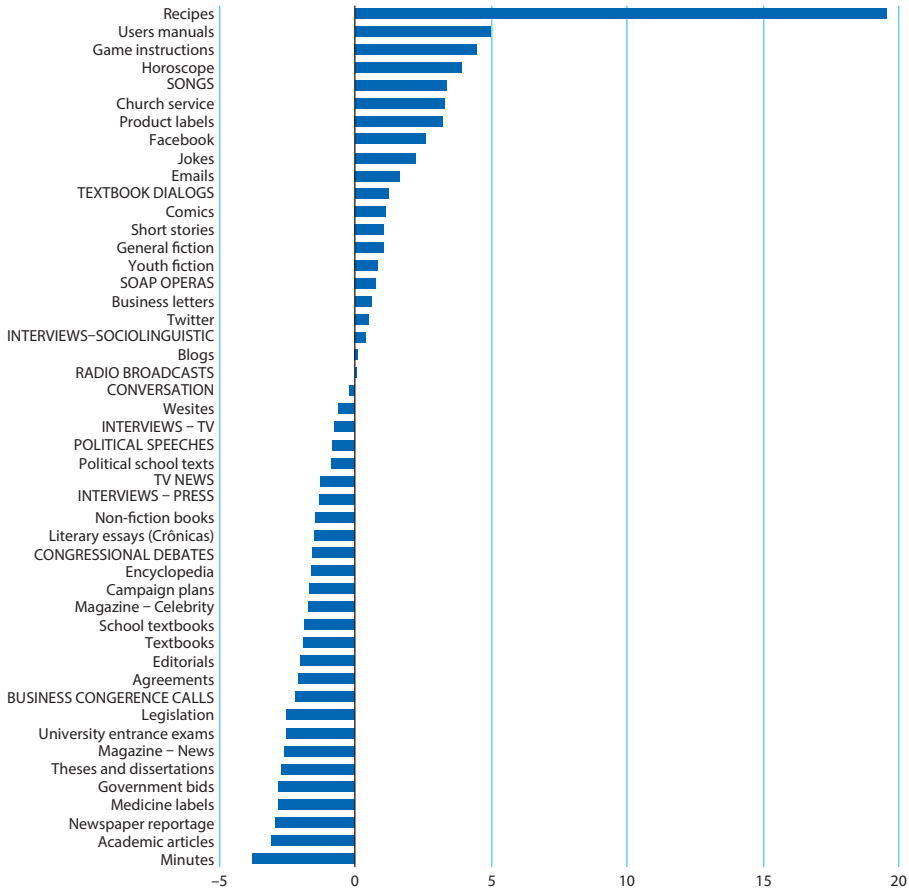


Figure 5. Dimension 4: Directive discourse

Example (6) illustrates the use of imperatives, concrete nouns, and adverbs of manner in a recipe.

- (6) *Caipirinha*
1 limão-galego
1 colher (sopa) de açúcar
2 cubos de gelo triturados

4 colheres (sopa) de pinga

Modo de Preparo: Lave o limão, corte em quatro pedaços. Coloque em um copo baixo e largo, junte açúcar e, com um socador, amasse até liberar todo o suco. Acrescente gelo triturado, pinga, misture bem e sirva.

Translation:

Capirinha

1 lime

1 tablespoon sugar

2 crushed ice cubes

4 tablespoons *cachaça*

Preparation: Wash the lime, cut it in four wedges. Put in a low wide glass, add the sugar, and with a crusher, mash until the juice from the lime has been squeezed out. Add the crushed ice, *cachaça*, mix well and serve.

The fact that medicine labels (*bulas*) are not marked for this dimension runs counter to expectations, as experience with such texts would dictate that they are directive. However, both the present subjunctive and the imperative – the main features marking directiveness on Dimension 4 – have low means in this register, respectively 4.1 and 1.2 per thousand words, compared to the most marked registers – namely, 39.3 and 35.3 for recipes and 21.1 and 14.3 for owner's manuals. Compared to the corpus as a whole, *bulas* are below average with respect to the subjunctive but above average in imperatives (the corpus means are respectively 2.05 and 3.5; see Table A1). A close examination of individual texts shows that directiveness in medicine labels in Brazilian Portuguese is typically expressed by features such as the infinitive form, the modal *dever* (usually in the negative), the adverb *somente* (only), and directly by numbers indicating the recommended dosage levels. The sample in Example (7) illustrates the use of these features in a *bula*.

- (7) *Dipirona não deve ser administrada em altas doses ou por períodos prolongados, sem controle médico. POSOLOGIA E ADMINISTRAÇÃO. Criança de 5,5 a 7,5 Kg: 0,1 à 0,2 ml - somente intramuscular. Criança de 8 a 10 Kg: 0,1 à 0,3 ml - somente intramuscular. (...) Adultos e adolescentes acima de 15 anos: 2 a 5 ml - IM ou IV. (...) Doses maiores, somente a critério médico. Aplicar a injeção endovenosa lentamente, 1 ml/minuto. Não misturar medicamentos na mesma seringa. PRECAUÇÕES. O uso de Dipirona deve ser evitado nos três primeiros meses e nas últimas 6 semanas da gestação e, mesmo fora destes períodos, somente administrar em gestantes em casos de extrema necessidade.*

Translation:

Dipyrone should not be administered in large doses or for prolonged periods without medical supervision. DOSAGE AND ADMINISTRATION. Children 5.5 to 7.5 kg: 0.1 to 0.2 ml – intramuscularly only. Children 8 to 10 kg: 0.1 to 0.3 ml – intramuscularly only. (...) Adults and adolescents over

15 years: 2 to 5 ml – IM or IV. (...) Larger doses, only by prescription. Apply intravenous injection slowly, 1 ml/minute. Do not mix medications in the same syringe. PRECAUTIONS. The use of dipyrone should be avoided in the first three months and the last 6 weeks of pregnancy, and even outside of these periods, only use in pregnant women in cases of extreme necessity.

Factor 5 has 15 features (Table 7), with the main ones being the future subjunctive and the present future indicative moods – both of which mark a time orientation toward the future. *Ou* (or) coordination can link both clauses and phrases, creating sequences of independent clauses or phrasal elements. The modal *dever* can be used as an obligation or a prediction modal, and *poder* can act as a prediction, ability, or permission modal. Subordinating conjunctions create dependent clauses, which in turn are attached to the main clause in different ways, such as through condition, cause, or comparison relationships. Probability adverbs mark the likelihood of an action or state. Phrasal coordination enables the linking of nouns and adjectives in phrases. Relational adjectives are classifiers that ‘have little descriptive content’ (Biber et al. 1999, p. 508). In contrast, the main features on the negative pole of the factor mark a time orientation toward the past: the

Table 7. Dimension 5: Future versus past time orientation

Feature [label]	Loading
Verbs: Future subjunctive mood [vbsubfut]	.616
Conjunctions: Coordinating (<i>ou</i>) [cjou]	.611
Verbs: Future present tense [vbfutpres]	.513
Modals: <i>Dever</i> [mddever]	.474
Modals: <i>Poder</i> [mdpoder]	.426
Subordinating (conditional) clause [cjcond]	.390
Adverbs: Likelihood [advlikl]	.389
Conjunctions: Coordinating (phrasal) [cjcoorphr]	.322
(Adjectives: Relational [adjrela])	.303)
Nouns: Place [nplac]	–.301
Verbs: Past subjunctive mood [vbsubpast]	–.308
(Articles: Indefinite [artindef])	–.320)
Adjectives: Affiliative [adjaffi]	–.355
Verbs: Imperfect [vbimperf]	–.375
Verbs: Past indicative tense [vbpast]	–.554

past indicative and the past imperfect tenses as well as the past subjunctive mood. In addition to these, affiliative adjectives express local and national designations, indefinite articles narrow down 'the reference to a single member of a class' (Biber et al. 1999, p. 70), and place nouns are used to refer to particular places in the discourse. These two poles basically mark a distinction between future and past; therefore, the interpretive label 'Future versus past time orientation' is used for Dimension 5.

The distribution of registers on Dimension 5 is shown in Figure 6. At the top of the chart are the registers using primarily the future tense in conjunction with *dever* and *poder* modals to express conditions, ability, and obligations.

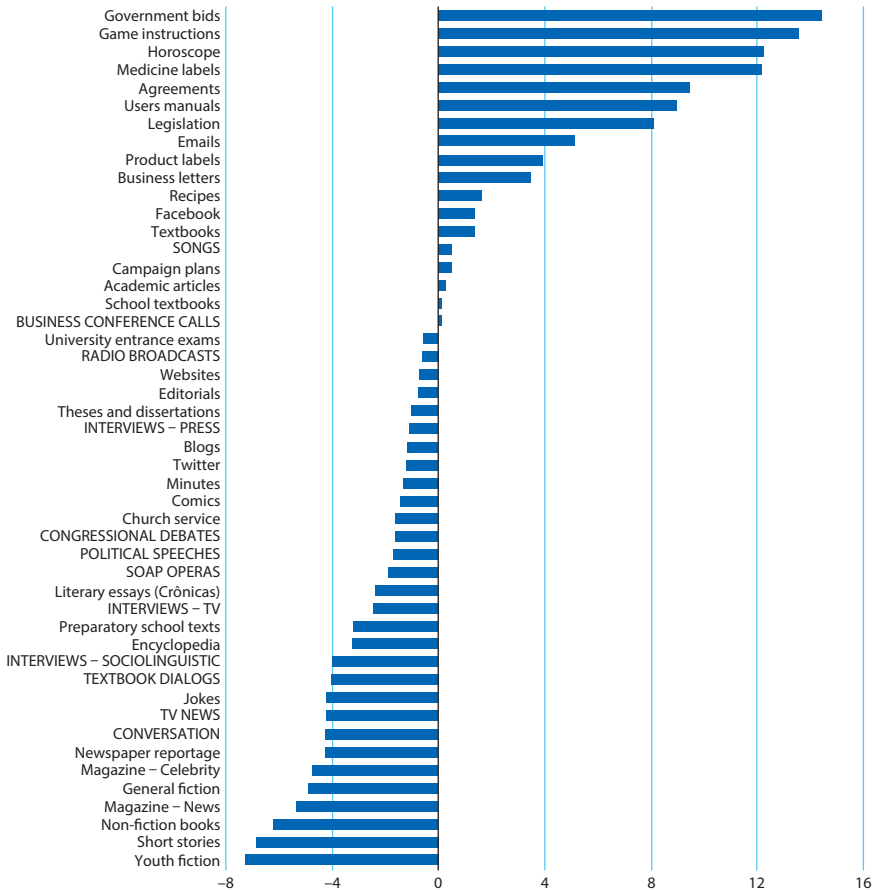


Figure 6. Dimension 5: Future versus past time orientation

Example (8) illustrates the use of such linguistic characteristics, in addition to *ou* conjunctions and phrasal coordination, in a government bid solicitation.

- (8) *A proponente que chegar atrasada será desclassificada. A presente licitação reger-se-á pelas normas e procedimentos do Regulamento Interno de Licitações e Contratos, e pelo presente instrumento convocatório. Participação: Poderão participar desta licitação empresas que comprovem estar devidamente regularizadas perante os órgãos públicos competentes (...). Não poderão participar da presente licitação: Empresas suspensas de licitar ou contratar com o [Government office] (...).*

Translation:

All late bidders will be disqualified. The present solicitation shall be governed by the provisions of the Internal Bids and Contracts Code and by the present solicitation instrument. Participation: Participation shall be granted to all businesses that demonstrate to be in good standing with the relevant public offices (...). The following shall not be able to participate in the current bid: All businesses that have been suspended from bidding or contracting with [Government office] (...).

At the other extreme are registers that rely on past tense forms to mark narrative concerns – primarily fiction registers, but also including other forms of storytelling, such as magazine news, newspaper reportage, conversation, and jokes. Example (9) illustrates the use of past tense forms in an excerpt from a youth fiction story.

- (9) *A Marcella vivia me protegendo, desde pequeno. Talvez porque, quando eu era bem criança, tive bronquite alérgica. Nem lembro bem como era, mas dizem que eu tossia tanto que até tinham medo de que eu botasse o pulmão pra fora. Desde então, ela cuidava de mim. Sempre me ajudava nos trabalhos da escola.*

Translation:

Marcella would always protect me, ever since I was little. Maybe because when I was a small kid, I had allergic bronchitis. I don't remember what it was like, but they say I would cough so much that they were afraid I would cough my lungs out. Since then, she has always taken care of me. Always helped me with my homework.

Dimension 6 (Table 8) has in fact only one pole (the positive one) because all features loading on the negative pole have higher loadings on other factors. Rare personal pronouns in the object position include the formal and archaic (in Brazilian Portuguese) *vos* (second-person plural) and *los* (third-personal plural) as well as contractions between some of these pronouns and preposition, such as *convosco*, formed by joining the preposition *com* (with) with the pronoun *vos*. Possessive pronouns refer to the participants in the text, and second-person verb

forms are used to refer to an addressee; therefore, both are related to interactivity (Biber 1988, p. 105). Final subordinating conjunctions (e.g. *para que*, ‘so that’) are complex subordinators (Biber et al. 1999, p. 282) that mark purpose in dependent clauses (Quirk, Greenbaum, Leech, & Svartvik 1985, p. 1070). *Que* clauses controlled by a preposition perform a number of different functions, depending on the preposition attached to them; in general, these are considered markers of formal discourse. Third-person pronouns in the object position refer to human or non-human referents in the discourse. *Que* relative clauses, as previously mentioned, provide ways in which to make elaborated reference to discourse participants. Public verbs can ‘function as markers of indirect, reported speech’ (Biber 1988, p. 109). Finally, the modal *haver* can be used as a formal marker of obligation. Based on the functions shared by these co-occurring features, the interpretive label for this dimension is Reported Discourse.

Table 8. Dimension 6: Reported discourse

Feature [label]	Loading
Pronouns: Rare in object position [objprnrare]	.628
Verbs: Second person [vb2]	.466
Pronouns: Possessive [prnposs]	.424
Subordinating (final) clause [cjfinal]	.413
(<i>Que</i> clause controlled by preposition [clqueeprp])	.380)
Pronouns: Third person, object position [prn3obl]	.371
(Pronouns: Relative <i>que</i> [prnque])	.340)
Verbs: Public [vbpubl]	.327
Modals: <i>Haver que/haver de</i> (have to, ought to) [mdhaver]	.311
(Adjectives: Evaluative [adjeval])	–.318)
(Yes or no question [qsyn])	–.330)
(Adverbs: Amplifier [advampl])	–.340)
(Adverbs: Intensity [advints])	–.341)

The distribution of registers on Dimension 6 is plotted in Figure 7, which shows that the most marked register is church services, followed by fiction and different story-telling registers in a distant second place. All of these registers rely on public verbs to report speech, but Order of Mass texts also make dense use of rare oblique pronouns and second-person verb forms (which are very infrequent in the other registers), thereby boosting their dimension scores.

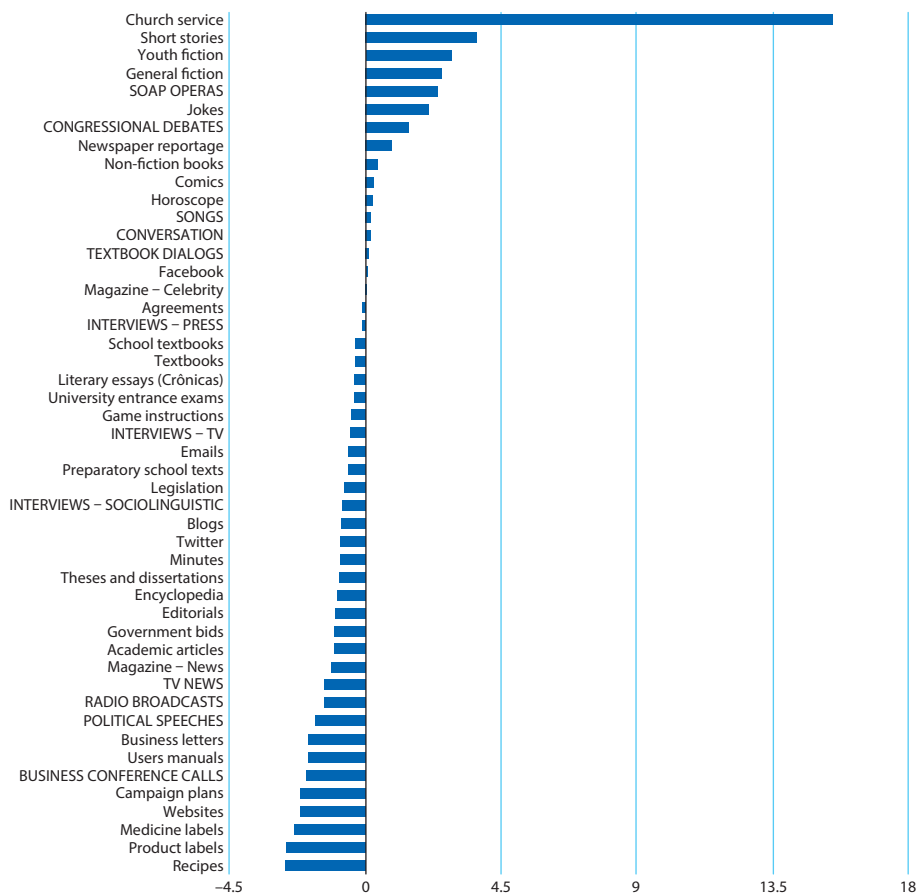


Figure 7. Dimension 6: Reported discourse

Example (10) is an excerpt of the Eucharistic prayer from an Order of Mass leaflet that illustrates the frequent use of second-person verb forms, normally in the imperative, in addition to public verbs, possessive pronouns (also known as possessive determiners in English grammar), and the archaic *vos* object pronoun.

- (10) *Ele tomou o cálice em suas mãos, deu graças novamente, e o deu a seus discípulos, dizendo: Tomai, todos, e bebei: este é o cálice do meu sangue, o sangue da nova e eterna aliança, que será derramado por vós e por todos, para remissão dos pecados. Fazei isto em memória de mim. Eis o mistério da fé! Todas as vezes que comemos deste pão e bebemos deste cálice, anunciamos, Senhor, a vossa morte, enquanto esperamos a vossa vinda!*

Translation:

Jesus took the chalice, and gave it to the disciples, and said, ‘Take this, all of you, and drink from it; for this is the chalice of my blood, the blood of

the new and everlasting covenant, which will be poured out for you and for many for the forgiveness of sins. Do this in memory of me. The mystery of faith! When we eat this Bread and drink this Cup, we proclaim your Death, O Lord, until you come again.

To summarize, the following dimensions were identified for Brazilian Portuguese:

1. Oral versus Literate Discourse
2. Argumentation
3. Involved versus Informational Production
4. Directive discourse
5. Future versus Past Orientation
6. Reported Discourse

As we used Promax rotation in the factor analysis, a residual amount of correlation remains between the factors, as shown in Table A4. The largest amount of correlation is between factors 1 and 3, which is understandable given the fact that both are versions of the same chief distinction between orality and literacy. As discussed, Dimension 1 represents this parameter more closely whereas Dimension 3 is a more specialized version that captures involvement and information more directly. As mentioned, these correlations fall within the normal range for MD studies of entire languages.

The F-test (ANOVA) performed on the mean dimension scores on each dimension indicates whether significant differences exist among the registers with respect to their mean dimension scores. In addition, the R^2 statistic ‘measures the percentage of the variance among dimension scores that can be predicted by knowing the register categories’ (Biber 1995, p. 119). Both statistics are shown in Table 9. The ANOVA results indicate that the differences among the registers are significant on all dimensions; the R^2 values suggest that five dimensions (all except Dimension 2) can be considered strong predictors of register differences

Table 9. ANOVAs and R^2 statistics

Dimension	F	p	df	R^2
1	97.037	.000	46	83.3%
2	19.926	.000	46	48.1%
3	101.521	.000	46	83.9%
4	57.961	.000	46	73.5%
5	4.561	.000	46	67.6%
6	43.525	.000	46	69.2%

in Brazilian Portuguese. These values are comparable to previous studies, such as for English (Biber 1988, p. 182), in which R^2 values ranged from 16.9 to 84.3; Tuvaluan, with values from 30 to 70; Korean, 5.5 to 61.5; and Somali, 19.4 to 90.9 (all from Biber 1995).

4. Discussion and conclusion

The dimensions found in this study have both similarities and differences with previous MD studies covering the scope of an entire national language. In terms of the similarities, all the dimensions identified for Brazilian Portuguese have correlates in MD analyses of other languages or varieties. Previous language-wide MD studies have all identified a dimension that reflects the distinction between oral and literate discourse, generally as the first and thus main factor. In English, it was initially called 'Interactive versus Edited Text' (Biber 1986) but was later renamed to 'Involved versus Informational Production' (Biber 1988). In Nukulaelae Tuvaluan, it is 'Interpersonal versus Information Reference' (Dimension 2) (Biber 1995), in Korean it is 'On-line Interaction versus Planned Exposition' (Dimension 1) (Biber 1995), in Somali it is 'Structural Elaboration: Involvement versus Exposition' (Dimension 1) (Biber 1995), and in Spanish it is 'Oral versus Literate Discourse' (Biber et al. 2006; Biber & Tracy-Ventura 2007). Both of our Dimensions 1 and 3 reflect this potentially universal dimension. Our Dimension 2 (Argumentation) occurs in English (Dimension 4) and in Somali in Dimensions 3 and 6 (Biber 1995). Our Dimension 4 (Directive Discourse) appears in Biber's (2006) study of university language and in Friginal's (2009) research on call centers (in both as Dimension 2). As it turns out, our Dimension 5 (Future versus Past Orientation) has no direct parallel, as it marks a distinction between future and past, not present and past, which would be more natural for narrativity. Finally, our Dimension 6 (Reported Discourse) is found in the early MD analysis of English (Dimension 3, viz. 'Reported versus Immediate Style' (Biber 1986), Korean (Dimension 5), Somali (Dimension 3).

At the same time, two particular characteristics distinguish our dimensions from previous MD research. The first is that the distinction between oral versus literature and between involved versus informational, which is considered the same dimension despite having different labels for the same dimension, was identified for two different dimensions in our study: Dimensions 1 and 3. The inter-factor correlation between these two dimensions is the highest (.41) in this study, thereby reflecting their mutual relationship. In our study, Dimension 1 marks a contrast between conversation-like and learned registers: On the one extreme, simulated dialogs (songs, soap operas, electronic communications, etc.) are or pretend to be

informal, everyday modes of colloquial interaction; on the other extreme, expository writing requires tertiary education to be produced and circulates primarily in restricted settings, such as universities (academic prose, university preparation textbooks), the legal system (legislation), or medical care (drug labels). Although such texts might be read by students, consumers, or the public in general, their production circumstances normally require extensive formal learning (a university degree). Hence, Dimension 1 could be best seen as reflecting the oral versus literate opposition. In contrast, Dimension 3 marks a contrast between actual interactive discourse, usually in the form of real-time turn-based dialog, exemplified by sociolinguistic interviews and conversation, and information-laden discourse that does not require higher education to be produced, such as recipes, business letters, and product labels. Although some of these registers do require technical expertise to be produced (e.g. agreements and some product labels), they do not necessarily require university-level education, whereas academic prose does, for instance (note that academic prose registers are not particularly marked for this dimension.) Based on these considerations, the interpretive label ‘Involved versus Informational Production’ was preferred for Dimension 3.

In short, we consider that underlying both Dimensions 1 and 3 is the same basic distinction between ‘stereotypically spoken discourse – that is, conversation’ – and ‘stereotypically written discourse – that is, informational exposition’ (Biber 1995, p. 238). However, we argue that the context of production of the literate/informational registers sets the two dimensions apart in our case. Hence, although the terms *literate* and *informational* have been conflated in previous research, in our current study we felt the need to separate them and attach particular meanings to them in order to reflect the different production circumstances operating on the registers.

The other distinguishing characteristic found in this study is that it did not reveal a clear-cut narrative dimension. Brazilian Portuguese does have a narrative dimension, but in our study it surfaced in conjunction with ‘future orientation’ on Dimension 5 rather than with a ‘present orientation’. Our study identified past forms as marking fiction registers, as in other MD studies (e.g. Biber 1988; Biber et al. 2006), but at the same time, it identified future tenses as a strong component for signaling obligation and prediction. This lack of a standard narrative dimension might have been the result of our corpus design, in which the representation of literature (youth fiction, general fiction, and short stories) is considerably lower than in previous MD research. As Biber et al. (2006, p. 30) advised, ‘it is important to bear in mind the possible influence of differences in corpus design.’ The share of fiction in the written component of the corpora in previous MD research ranges from 7.5% in Spanish (Biber et al. 2006, p. 7) to 12.5% in Korean (Biber 1995, p. 91), 21.7% in English (Biber 1988, p. 67), and 6.3% in ours. Although the difference

between our study and the Spanish MD analysis of Biber et al. (2006) is not great, there is a large difference with respect to the word count of fiction texts as a share of the total written subcorpus, which is about 50% for the Spanish analysis and about 25% for ours. This difference in the weight of literary texts might have played a role, among other influences, in the rise of this mixed-narrative dimension in our study.

The completion of this analysis affords the opportunity to contrast the dimensions of two closely related languages – namely, Portuguese and Spanish (Biber et al. 2006; Biber & Tracy-Ventura 2007). Although both analyses identified the same number of dimensions (6) for each language, only two dimensions had counterparts in both languages: oral/literate discourse (Dimension 1 in both studies) and narrative discourse (Dimension 3 in Spanish and 5 here, with the previously discussed caveat). This suggests that, although the two languages share many linguistic features in common (notably the vocabulary), they use linguistic resources differently for communicative purposes. Again, a good deal of the mismatch might be attributable to differences in corpus design, as discussed herein. However, as Biber et al. (2006, p. 30) argued, we should not overstate this influence if the corpora ‘cover roughly the same range of registers, differing primarily in the relative weightings given to particular registers.’ More research is needed to assess the influence of different corpora on MD analysis to resolve this issue. Nevertheless, if the differences identified here hold in a more controlled comparison, it would lead us to conclude that the register differences between Portuguese and Spanish are real and extend far beyond corpus selection criteria.

Our study is unlike previous language-wide MD research in that it restricted the scope of the investigation to one variety of Portuguese, rather than attempting to comprise many different dialects. Although other studies, such as for English and Spanish, opted to represent different varieties, consideration of the marked syntactic and lexical differences among Brazilian, European, and African varieties of Portuguese (Berber Sardinha & São Bento Ferreira 2014; Castilho 2009) suggests that it would be prudent to focus on one single variety.

Future MD research should consider exploring other varieties of Portuguese to determine if there are indeed significant differences in the multidimensional space of register variation in the major dialects of Portuguese around the world.

Acknowledgments

The authors wish to thank both Fapesp (the São Paulo State Research Council) and CNPq (Brasília, DF) for the grants that supported the Dimensions of Register Variation in Brazilian Portuguese project. Tony Berber Sardinha acknowledges the continued support of CNPq (Brasília, DF) for his research.

References

- Azevedo, M. M. (2005). *Portuguese: A linguistic introduction*. Cambridge: CUP. DOI: 10.1017/CBO9780511755057
- Bacelar do Nascimento, M. F., Mendes, A., Antunes, S., & Pereira, L. (2014). The Reference Corpus of Contemporary Portuguese and related resources. In T. Berber Sardinha & T. São Bento Ferreira (Eds.), *Working with Portuguese corpora* (pp. 237–256). London: Bloomsbury.
- Bechara, E. (1999). *Moderna gramática portuguesa* [Modern Portuguese grammar] (37th Ed.). Rio de Janeiro, RJ: Lucerna.
- Berber Sardinha, T. (Ed.). (2005). *A língua portuguesa no computador* [The Portuguese language on computer]. Campinas, SP: Mercado de Letras/FAPESP.
- Berber Sardinha, T., & São Bento Ferreira, T. (2014). *Working with Portuguese corpora*. London: Bloomsbury.
- Berber Sardinha, T., São Bento Ferreira, T., & Teixeira, R. d. B. S. (2014). Lexical bundles in Brazilian Portuguese. In T. Berber Sardinha & T. São Bento Ferreira (Eds.), *Working with Portuguese corpora* (pp. 33–68). London: Bloomsbury.
- Besnier, N. (1988). The linguistic relationships of spoken and written Nukulaelae registers. *Language*, 64, 707–736. DOI: 10.2307/414565
- Biber, D. (1986). Spoken and written textual dimension in English: Resolving the contradictory findings. *Language*, 62, 384–414. DOI: 10.2307/414678
- Biber, D. (1988). *Variation across speech and writing*. Cambridge: CUP. DOI: 10.1017/CBO9780511621024
- Biber, D. (1995). *Dimensions of register variation: A cross-linguistic comparison*. Cambridge: CUP. DOI: 10.1017/CBO9780511519871
- Biber, D. (2006). *University language: A corpus-based study of spoken and written registers*. Amsterdam: John Benjamins. DOI: 10.1075/scl.23
- Biber, D., Davies, M., Jones, J. K., & Tracy-Ventura, N. (2006). Spoken and written register variation in Spanish: A multi-dimensional analysis. *Corpora*, 1(1), 1–37. DOI: 10.3366/cor.2006.1.1.1
- Biber, D., & Hared, M. (1994). Linguistic correlates of the transition to literacy in Somali: Language adaptation in six press registers. In D. Biber & E. Finegan (Eds.), *Sociolinguistic perspectives on register* (pp. 182–216). Oxford: OUP.
- Biber, D., Johansson, S., Leech, G., Conrad, S., & Finegan, E. (1999). *The Longman grammar of spoken and written English*. London: Longman.
- Biber, D., & Tracy-Ventura, N. (2007). Dimensions of register variation in Spanish. In G. Parodi (Ed.), *Working with Spanish corpora* (pp. 54–89). London: Continuum.
- Bick, E. (2014). PALAVRAS, a constraint grammar-based parsing system for Portuguese. In T. Berber Sardinha, & T. São Bento Ferreira (Eds.), *Working with Portuguese corpora* (pp. 279–302). London: Bloomsbury.
- Castilho, A. T. d. (2009). Portuguese. In K. Brown & S. Ogilvie (Eds.), *Concise encyclopedia of languages of the world* (pp. 883–885). Oxford: Elsevier.
- Castilho, A. T. d. (Ed.). (1989). *Português culto falado no Brasil* [Standard spoken Portuguese in Brazil]. Campinas, SP: Editora da Unicamp.
- Crossley, S., & Louwerse, M. M. (2007). Multi-dimensional register classification using bi-grams. *International Journal of Corpus Linguistics*, 12(4), 453–478. DOI: 10.1075/ijcl.12.4.02cro

- Cunha, C. (2001). *Nova gramática do português contemporâneo* [New grammar of contemporary Portuguese]. Rio de Janeiro, RJ: Nova Fronteira.
- de Mönnink, I. M., Brom, N., & Oostdijk, N. H. J. (2003). Using the MF/MD method for automatic text classification. In S. Granger & S. Petch Tyson (Eds.), *Extending the scope of corpus based research: New applications new challenges* (pp. 15–25). Amsterdam: Rodopi.
- Didriksen, T., Berber Sardinha, T., & Bick, E. (2013). *CG3-IDE Software*. Aarhus; Odense; São Paulo, SP: FAPESP/CEPRIL/GrammarSoft. Retrieved from <http://visl.sdu.dk/download/vislcg3/>
- Friginal, E. (2009). *The language of outsourced call centers: A corpus-based study of cross-cultural interaction*. Amsterdam: John Benjamins. DOI: 10.1075/scl.34
- Ilari, R. (1991). *Gramática do português falado: Níveis de análise linguística* [Grammar of spoken Portuguese: Levels of linguistic analysis]. Campinas, SP: Editora da Unicamp.
- Kauffmann, C. (2005). *O Corpus do Jornal: Variação linguística, gêneros e dimensões da imprensa diária escrita* [The Newspaper Corpus: Linguistic variation, genres, and dimensions in the daily press] (Unpublished master's thesis). São Paulo Catholic University, São Paulo, Brazil.
- Kilgarriff, A., Jakubíček, M., Pomikalek, J., Berber Sardinha, T., & Whitelock, P. (2014). PtTenTen: A corpus for Portuguese lexicography. In T. Berber Sardinha & T. São Bento Ferreira (Eds.), *Working with Portuguese corpora* (pp. 111–130). London: Bloomsbury.
- Kim, Y.-J., & Biber, D. (1994). A corpus-based analysis of register variation in Korean. In D. Biber, & E. Finegan (Eds.), *Sociolinguistic perspectives on register* (pp. 157–181). Oxford: OUP.
- Lamb, W. (2008). *Scottish Gaelic speech and writing: Register variation in an endangered language*. Belfast: Cló Ollscoil na Banríona.
- Lee, D. Y. W. (1999). *Modelling variation in spoken and written language: The multi-dimensional approach revisited*. (Unpublished doctoral dissertation). Lancaster University, Lancaster, UK.
- Moura Neves, M. H. d. (2000). *Gramática de usos do português* [Portuguese usage grammar]. São Paulo, SP: Editora Unesp.
- Parodi, G. (2007). Variation across registers in Spanish: Exploring the El-Grial PUCV Corpus. In G. Parodi (Ed.), *Working with Spanish corpora* (pp. 11–53). London: Continuum.
- Preti, D. (Ed.). (2005). *O discurso oral culto* [Standard oral discourse] (3rd Ed.). São Paulo, SP: Humanitas.
- Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1985). *A comprehensive grammar of the English language*. London: Longman.
- Souza e Silva, M. C. P. d., & Brait, B. (Eds.). (2013). *Texto ou discurso?* [Text or discourse?]. São Paulo, SP: Contexto.
- Thomas, E. W. (1969). *The syntax of spoken Brazilian Portuguese*. Nashville, TN: Vanderbilt University Press.
- Whitlam, J. (2010). *Modern Brazilian Portuguese grammar*. New York, NY: Routledge.

Appendix

Features loading on final factor analysis

- 1 Adjectives: Affiliative
- 2 Adjectives: Attributive position
- 3 Adjectives: Attributive, pre-modifying
- 4 Adjectives: Evaluative
- 5 Adjectives: Predicative position
- 6 Adjectives: Relational
- 7 Adjectives: Topical
- 8 Adverbs: *Não*
- 9 Adverbs: Amplifier
- 10 Adverbs: Comparative
- 11 Adverbs: Emphatic
- 12 Adverbs: Hedge
- 13 Adverbs: Intensity
- 14 Adverbs: Likelihood
- 15 Adverbs: Manner
- 16 Adverbs: Negative, except *não*
- 17 Adverbs: Place
- 18 Adverbs: Time
- 19 Articles: Definite
- 20 Articles: Indefinite
- 21 Clause types: Infinitive clause controlled by adjective
- 22 Clause types: Infinitive clause controlled by ease or difficulty adjective
- 23 Clause types: Infinitive clause controlled by preposition
- 24 Clause types: *Que* clause controlled by adjective (stance)
- 25 Clause types: *Que* clause controlled by adverb
- 26 Clause types: *Que* clause controlled by noun
- 27 Clause types: *Que* clause controlled by preposition
- 28 Clause types: *Que* clause controlled by verb in indicative mood
- 29 Clause types: *Que* or infinitive clause controlled by noun (stance)
- 30 Clause types: Reduced progressive clause
- 31 Clause types: Subordinating (conditional)
- 32 Clause types: Subordinating (final)
- 33 Conjunctions: Coordinating (*ou*)
- 34 Conjunctions: Coordinating (adversative)
- 35 Conjunctions: Coordinating (clausal)
- 36 Conjunctions: Coordinating (conclusive)
- 37 Conjunctions: Coordinating (phrasal)
- 38 Modals: *Dever*
- 39 Modals: *Haver que/haver de*
- 40 Modals: *Poder*
- 41 Modals: *Precisar*
- 42 Modals: *Ter que/ter de*
- 43 Nouns: Abstract

- 44 Nouns: Cognition
- 45 Nouns: Compound
- 46 Nouns: Concrete
- 47 Nouns: Nominalization in subject position
- 48 Nouns: Place
- 49 Other features of verb or noun phrases: Agentless passive
- 50 Other: Contractions
- 51 Other: Discourse marker
- 52 Other: Subject omission
- 53 Prepositions: All
- 54 Pronouns: Demonstrative
- 55 Pronouns: First person singular, in subject position
- 56 Pronouns: First person, object position
- 57 Pronouns: Nominal in subject position
- 58 Pronouns: Possessive
- 59 Pronouns: Quantifier
- 60 Pronouns: Rare in object position
- 61 Pronouns: Relative *qual* or *cujo*
- 62 Pronouns: Relative *que*
- 63 Pronouns: Second person singular, in subject position
- 64 Pronouns: Second person, object position
- 65 Pronouns: Third person plural, in subject position
- 66 Pronouns: Third person singular, in subject position
- 67 Pronouns: Third person, object position
- 68 Questions: *QU* questions
- 69 Questions: Tag questions
- 70 Questions: Yes or No question
- 71 Verbs: *Ir* future
- 72 Verbs: Action
- 73 Verbs: Communication
- 74 Verbs: Facilitation
- 75 Verbs: First person
- 76 Verbs: Future present tense
- 77 Verbs: Future preterite tense
- 78 Verbs: Future subjunctive mood
- 79 Verbs: Gerund form, all
- 80 Verbs: Imperative mood
- 81 Verbs: Imperfect
- 82 Verbs: Infinitive
- 83 Verbs: Mental
- 84 Verbs: Past indicative tense
- 85 Verbs: Past participle
- 86 Verbs: Past subjunctive mood
- 87 Verbs: Present subjunctive mood
- 88 Verbs: Private
- 89 Verbs: Progressive preceded by infinitive

- 90 Verbs: Public
 91 Verbs: Second person
 92 Vocabulary distribution: Average word length
 93 Vocabulary distribution: Type-token ratio

Table A1. Descriptive statistics

#	Feature:	Min	Max	Mean	Std. Dev.
1	Adjectives: Affiliative	.00	17.54	1.46	2.51
2	Adjectives: All	9.89	157.35	60.50	22.19
3	Adjectives: Attributive position	1.75	107.00	37.66	18.66
4	Adjectives: Attributive, post-modifying	.00	97.28	27.18	15.86
5	Adjectives: Attributive, pre-modifying	.00	31.73	9.37	5.34
6	Adjectives: Augmentative	.00	8.99	.58	.95
7	Adjectives: Color	.00	24.66	.57	1.72
8	Adjectives: Evaluative	.00	27.40	5.57	4.27
9	Adjectives: Except evaluative	.00	70.53	25.38	12.46
10	Adjectives: Predicative position	.00	35.34	8.82	5.27
11	Adjectives: Relational	.00	43.66	10.98	6.23
12	Adjectives: Size	.00	24.90	3.50	2.97
13	Adjectives: Time	.00	16.98	2.57	2.50
14	Adjectives: Topical	.00	38.80	6.31	6.19
15	Adverbs: <i>Não</i>	.00	95.87	7.97	6.73
16	Adverbs: All	2.58	210.53	59.63	31.88
17	Adverbs: Amplifier	.00	34.92	7.04	5.05
18	Adverbs: Attitudinal	.00	14.93	1.59	1.71
19	Adverbs: Comparative	.00	45.39	9.38	5.15
20	Adverbs: Compound	.00	31.75	7.80	4.48
21	Adverbs: Downtoner	.00	10.66	1.28	1.51
22	Adverbs: Emphatic	.00	27.15	5.06	3.95
23	Adverbs: Exc. time, manner and place	.00	125.94	25.37	14.46
24	Adverbs: Factive	.00	109.02	13.93	10.77
25	Adverbs: Hedge	.00	12.11	.88	1.30
26	Adverbs: Intensity	.00	31.10	6.59	4.91
27	Adverbs: Likelihood	.00	15.57	1.51	1.67
28	Adverbs: Long (10 letters or more)	.00	62.75	9.82	10.67
29	Adverbs: Manner	.00	17.58	2.06	2.64

(Continued)

Table A1. (Continued)

#	Feature:	Min	Max	Mean	Std. Dev.
30	Adverbs: Negative, except <i>não</i>	.00	6.59	.42	.84
31	Adverbs: Non-factual	.00	14.96	1.75	1.80
32	Adverbs: Place	.00	42.33	4.08	5.01
33	Adverbs: Time	.00	32.97	5.58	4.49
34	Articles: Definite	27.34	192.88	113.62	28.08
35	Articles: Indefinite	.00	48.16	14.02	7.84
36	Conjunctions: Coordinating (<i>ou</i>)	.00	26.69	3.39	4.22
37	Conjunctions: Coordinating (additive)	1.78	73.30	31.50	10.05
38	Conjunctions: Coord. (adversative)	.00	17.01	2.34	2.32
39	Conjunctions: Coordinating (all)	7.46	83.14	34.05	10.57
40	Conjunctions: Coordinating (clausal)	.00	40.86	11.37	6.04
41	Conjunctions: Coord. (conclusive)	.00	14.26	1.64	2.05
42	Conjunctions: Coordinating (phrasal)	.00	41.57	6.30	5.36
43	CU clause cntrld. by verb	.00	17.58	1.55	1.60
44	Infinitive clause cntrld. by adjective	.00	13.61	.58	1.06
45	Inf. clause cntrld. by affective adjective	.00	4.04	.01	.16
46	Inf. clause cntrld. by attitude adjective	.00	8.33	.22	.62
47	Inf. clause cntrld. by attitudinal noun	.00	7.80	.20	.55
48	Inf. clause cntrld. by causative verb	.00	5.53	.20	.53
49	Inf. clause cntrld. by causative verb	.00	2.27	.04	.19
50	Inf. clause cntrld. by certainty adjective	.00	1.73	.01	.08
51	Inf. clause cntrld. by cognitive verb	.00	6.94	.06	.34
52	Inf. clause cntrld. by desire verb	.00	3.96	.07	.30
53	Inf. cl. cntrld. by ease or difficulty adj.	.00	8.77	.23	.62
54	Inf. clause cntrld. by evaluation adj.	.00	4.54	.15	.43
55	Inf. clause cntrld. by factive adjective	.00	8.33	.06	.47
56	Inf. clause cntrld. by factual noun	.00	3.28	.12	.34
57	Inf. cl. cntrld. by likelihood adjective	.00	4.87	.10	.35
58	Inf. clause cntrld. by likelihood noun	.00	4.53	.12	.41
59	Inf. clause cntrld. by non-factual noun	.00	1.87	.04	.17
60	Infinitive clause cntrld. by preposition	.00	31.97	9.10	4.66
61	Infinitive clause cntrld. by probability verb	.00	1.70	.01	.08
62	Infinitive clause cntrld. by speech verb	.00	1.77	.02	.10

(Continued)

#	Feature:	Min	Max	Mean	Std. Dev.
63	Inf. cl. cntrld. by willingness adjective	.00	4.16	.15	.42
64	Modals: All	.00	30.54	5.75	4.51
65	Modals: <i>Conseguir</i>	.00	6.54	.31	.71
66	Modals: <i>Dever</i>	.00	16.38	1.21	1.94
67	Modals: <i>Haver que/de</i>	.00	2.36	.05	.23
68	Modals: Obligation	.00	17.11	2.26	2.37
69	Modals: <i>Parecer</i>	.00	3.79	.10	.37
70	Modals: <i>Poder</i>	.00	21.85	3.07	2.89
71	Modals: <i>Precisar</i>	.00	7.14	.34	.79
72	Modals: <i>Ter que/de</i>	.00	7.09	.66	1.08
73	Nouns: Abstract	1.78	142.86	52.10	27.21
74	Nouns: All, except nominalizations	56.64	354.76	159.65	37.52
75	Nouns: Animate	.00	54.15	15.57	10.45
76	Nouns: Cognition	.00	47.09	6.85	5.76
77	Nouns: Compound	.00	66.98	16.08	10.46
78	Nouns: Concrete	1.42	84.54	20.95	12.25
79	Nouns: In subject position	.00	62.39	25.82	10.36
80	Nouns: Institution	.00	30.76	3.08	3.63
81	Nouns: Nominalization	.00	156.11	46.10	25.27
82	Nouns: Nominaliz. in subject position	.00	35.05	6.95	5.16
83	Nouns: Place	.00	61.96	7.97	6.82
84	Nouns: Proper	.00	735.65	55.90	60.37
85	Nouns: Quantity	.00	69.92	14.04	8.89
86	Nouns: Technical	.00	50.19	3.84	4.89
87	Other: <i>Por</i> passive	.00	6.27	.40	.80
88	Other: <i>Se</i> passive	.00	6.14	.43	.80
89	Other: Agentless passive	.00	14.32	1.95	2.38
90	Other: Passive postnominal modifier	.00	25.81	4.20	3.84
91	Other: Contractions	.00	36.84	1.33	4.30
92	Other: Discourse marker	.00	56.13	5.41	5.97
93	Other: Focus marker	.00	11.70	.76	1.26
94	Other: Subject omission	.00	87.75	13.57	11.07
95	Prepositions: All	49.15	316.47	147.23	33.79
96	Pronouns: Demonstrative	.00	51.44	11.51	6.94

(Continued)

Table A1. (Continued)

#	Feature:	Min	Max	Mean	Std. Dev.
97	Pronouns: 1st p. plural, in subject pos.	.00	18.84	.56	1.52
98	Pronouns: 1st p. sing., in subject pos.	.00	47.86	3.09	5.94
99	Pronouns: 1st person, object position	.00	99.79	2.11	5.29
100	Pronouns: Nominal in subject position	.00	7.56	.64	1.06
101	Pronouns: Possessive	.00	56.39	10.66	8.98
102	Pronouns: Quantifier	.00	63.35	18.84	9.98
103	Pronouns: Rare in object position	.00	13.72	.88	1.66
104	Pronouns: Relative <i>qual</i> or <i>cujo</i>	.00	7.91	.60	1.03
105	Pronouns: Relative <i>que</i>	.00	32.32	10.21	5.50
106	Pronouns: Relative prec. by preposition	.00	9.52	1.05	1.30
107	Pronouns: 2nd p. plural, subject pos.	.00	6.96	.24	.67
108	Pronouns: 2nd p sing., subject pos.	.00	30.40	2.32	4.00
109	Pronouns: 2nd p., object position	.00	86.40	5.06	8.81
110	Pronouns: 3rd p. plural, subject pos.	.00	9.89	.56	1.10
111	Pronouns: 3rd p. sing., subject pos.	.00	43.20	2.69	4.41
112	Pronouns: 3rd p., object position	.00	41.35	7.37	5.37
113	<i>Que</i> clause cntrld. by adjective	.00	5.38	.45	.76
114	<i>Que</i> clause cntrld. by adjective	.00	7.03	1.24	1.28
115	<i>Que</i> clause cntrld. by adj. of certainty	.00	3.49	.02	.17
116	<i>Que</i> cl. cntrld. by adj. of ease or diffc.	.00	2.32	.03	.16
117	<i>Que</i> cl. cntrld. by adj. of evaluation	.00	3.14	.08	.28
118	<i>Que</i> clause cntrld. by adverb	.00	21.48	.53	1.25
119	<i>Que</i> clause cntrld. by attitude adjective	.00	2.21	.02	.12
120	<i>Que</i> clause cntrld. by attitude noun	.00	2.21	.03	.16
121	<i>Que</i> clause cntrld. by cognitive verb	.00	6.24	.20	.51
122	<i>Que</i> clause cntrld. by desire verb	.00	3.97	.04	.20
123	<i>Que</i> clause cntrld. by factive adjective	.00	4.30	.05	.26
124	<i>Que</i> clause cntrld. by factual noun	.00	3.55	.08	.29
125	<i>Que</i> cl. cntrld. by likelihood adjective	.00	2.32	.03	.19
126	<i>Que</i> clause cntrld. by likelihood noun	.00	5.10	.13	.40
127	<i>Que</i> clause cntrld. by non-factual noun	.00	3.60	.02	.17
128	<i>Que</i> clause cntrld. by noun	.00	18.48	5.85	3.63
129	<i>Que</i> clause cntrld. by preposition	.00	8.08	1.05	1.28
130	<i>Que</i> clause cntrld. by probability verb	.00	1.02	.00	.04

(Continued)

#	Feature:	Min	Max	Mean	Std. Dev.
131	<i>Que</i> clause cntrld. by speech verb	.00	8.33	.23	.67
132	<i>Que</i> clause cntrld. by verb	.00	24.34	4.95	4.36
133	<i>Que</i> cl. cntrld. by indicative verb	.00	24.34	3.76	3.87
134	<i>Que</i> clause cntrld. by subjunctive	.00	1.68	.01	.09
135	<i>Que</i> or inf. clause cntrld. by adjective	.00	16.67	.48	1.08
136	<i>Que</i> or infinitive clause cntrld. by noun	.00	8.50	.73	1.02
137	<i>Que</i> or infinitive clause cntrld. by verb	.00	8.93	.87	1.23
138	Questions: <i>QU</i> questions	.00	18.10	1.36	2.49
139	Questions: Tag questions	.00	40.09	.98	4.17
140	Questions: Yes or No question	.00	56.07	3.70	7.63
141	Reduced progressive clause	.00	21.43	3.26	2.73
142	Sub. cl., exc. causal /conc./cond.	.00	11.88	2.13	1.77
143	Subordinating (all)	.00	54.70	13.99	8.94
144	Subordinating (causal)	.00	39.06	9.92	6.99
145	Subordinating (concessive)	.00	4.25	.31	.59
146	Subordinating (conditional)	.00	13.99	1.63	1.98
147	Subordinating (conformative)	.00	1.97	.04	.18
148	Subordinating (final)	.00	1.70	.01	.08
149	Subordinating (proportional)	.00	1.70	.01	.09
150	Subordinating (temporal)	.00	11.88	2.07	1.75
151	Verbs: <i>Ir</i> future	.00	33.84	2.73	3.96
152	Verbs: <i>Ser</i> or <i>estar</i>	.00	90.48	19.51	12.66
153	Verbs: Action	5.89	111.33	36.99	14.52
154	Verbs: All	25.18	364.36	156.15	46.02
155	Verbs: Aspectual	.00	13.16	.46	.94
156	Verbs: Auxiliary	.00	24.44	5.12	3.59
157	Verbs: Communication	.00	37.11	8.13	6.41
158	Verbs: Existence or relation	.00	111.23	43.69	19.36
159	Verbs: Facilitation	.00	23.42	3.81	2.98
160	Verbs: First person	.00	109.55	16.22	18.77
161	Verbs: Future present tense	.00	62.50	3.92	6.18
162	Verbs: Future preterit tense	.00	25.59	1.81	2.61
163	Verbs: Future subjunctive mood	.00	28.22	2.05	3.00
164	Verbs: Gerund form, all	.00	75.25	8.27	5.45
165	Verbs: Imperative mood	.00	62.79	3.48	6.98

(Continued)

Table A1. (Continued)

#	Feature:	Min	Max	Mean	Std. Dev.
166	Verbs: Imperfect	.00	87.64	8.32	11.82
167	Verbs: Indicative mood	.00	238.28	84.73	40.78
168	Verbs: Infinitive	.00	21.07	2.55	2.30
169	Verbs: Mental	.00	88.17	16.87	11.15
170	Verbs: Occurrence	.00	18.83	3.32	2.48
171	Verbs: Past indicative tense	.00	89.02	19.56	17.32
172	Verbs: Past participle	.00	62.89	18.63	10.68
173	Verbs: Past subjunctive mood	.00	21.32	1.02	1.67
174	Verbs: Perfect aspect	.00	100.47	38.53	17.83
175	Verbs: Personal infinitive	.00	19.16	2.13	2.15
176	Verbs: Pluperfect tense	.00	11.92	.34	1.01
177	Verbs: Present indicative tense	.00	214.84	53.62	32.47
178	Verbs: Present subjunctive mood	.00	71.30	6.31	7.53
179	Verbs: Private	.00	52.88	12.41	8.11
180	Verbs: Progressive	.00	6.05	.38	.72
181	Verbs: Progressive preceded by inf.	.00	22.56	.16	.84
182	Verbs: Public	.00	37.70	9.76	6.46
183	Verbs: Second person	.00	35.32	1.36	3.91
184	Verbs: Split auxiliary	.00	3.06	.08	.26
185	Verbs: Suasive	.00	27.14	2.66	2.52
186	Verbs: Third person	7.36	203.13	83.85	29.32
187	Vocabulary distr.: avg. clause length	.08	100.00	8.73	8.65
188	Vocabulary distr.: Average word length	3.62	6.57	4.78	.43
189	Vocabulary distr.: Type-token ratio	.17	.75	.53	.07
190	Vocabulary distribution: Word count	401.00	107911.00	5879.17	13256.77

Table A2. Factor pattern after promax rotation

Feature	F1	F2	F3	F4	F5	F6
Adjectives: Affiliative	-.34	.12	-.01	-.14	-.35	-.06
Adjectives: Attributive position	-.59	.21	-.12	-.08	.11	-.29
Adjectives: Attributive, pre-modifying	-.11	.17	-.36	-.23	-.01	-.21
Adjectives: Augmentative	.04	.01	-.12	-.12	-.01	-.27

(Continued)

Feature	F1	F2	F3	F4	F5	F6
Adjectives: Color	.02	-.05	-.01	.22	-.02	.05
Adjectives: Evaluative	.61	.23	-.16	-.07	.01	-.32
Adjectives: Predicative position	.46	.35	-.05	-.01	-.09	-.19
Adjectives: Relational	-.42	.25	.01	.03	.30	-.18
Adjectives: Size	-.24	.20	-.07	.20	-.14	-.21
Adjectives: Time	.00	.03	-.22	-.26	.05	-.10
Adjectives: Topical	-.51	.23	.09	-.20	.06	-.20
Adverbs: <i>Não</i> (no)	.67	.14	.19	-.12	.17	.03
Adverbs: Amplifier	.52	.29	-.04	-.03	-.16	-.34
Adverbs: Attitudinal	.22	.18	.06	.01	-.05	.00
Adverbs: Comparative	.20	.47	-.08	-.05	.00	-.21
Adverbs: Compound	.24	.20	.26	.22	-.03	-.05
Adverbs: Downtoner	.27	.20	.02	-.01	-.17	-.01
Adverbs: Emphatic	.48	.22	-.02	.23	-.10	-.11
Adverbs: Hedge	.22	.33	-.02	.01	-.09	.03
Adverbs: Intensity	.55	.26	-.06	.13	-.24	-.34
Adverbs: Likelihood	.08	.28	-.06	-.01	.39	.00
Adverbs: Manner	.37	-.01	.29	.34	-.02	-.18
Adverbs: Negative, except <i>não</i>	.32	.09	-.06	-.04	-.10	.07
Adverbs: Non-factual	-.24	.12	.04	-.07	.02	-.11
Adverbs: Place	.46	-.18	.46	-.03	.02	-.12
Adverbs: Time	.63	.11	-.05	-.04	-.15	-.09
Articles: Definite	-.74	.07	.00	-.12	-.16	.11
Articles: Indefinite	-.10	.33	.20	.21	-.32	-.02
CU clause controlled by verb	.13	.23	.06	-.02	.00	.16
Infinitive clause controlled by adjective	-.01	.45	-.07	.06	.08	-.01
Infinitive clause controlled by affective adjective	.01	.03	-.02	.00	-.02	.03
Infinitive clause controlled by by causative verb	.11	.21	.06	.07	.16	.10
Infinitive clause controlled by by cognitive verb	.19	.07	-.01	.01	.07	.05
Infinitive clause controlled by by desire verb	.17	.04	.05	.01	.06	.01
Infinitive clause controlled by by likelihood adjective	-.12	.24	.01	.02	.06	.01
Infinitive clause controlled by by probability verb	-.04	.04	-.01	.04	.04	.23
Infinitive clause controlled by causative verb	-.11	.11	.04	.08	.02	.13

(Continued)

Table A2. (Continued)

Feature	F1	F2	F3	F4	F5	F6
Infinitive clause controlled by certainty adjective	.07	.18	-.05	.04	.10	-.01
Infinitive clause cont. by ease or difficulty adjective	-.03	.33	-.04	.07	.06	-.01
Infinitive clause controlled by evaluation adjective	.09	.29	-.03	.02	.07	-.05
Infinitive clause controlled by factual noun	-.15	.10	.02	-.10	-.03	-.01
Infinitive clause controlled by likelihood noun	.00	.16	-.02	-.02	.08	-.01
Infinitive clause controlled by non-factual noun	-.08	.21	.15	.03	.11	.06
Infinitive clause controlled by preposition	.12	.40	-.13	.20	-.03	.10
Infinitive clause controlled by willingness adjective	-.08	.15	-.03	-.05	.04	.03
Que clause controlled by adjective	-.03	.38	-.02	-.11	-.03	-.05
Que clause controlled by adjective of certainty	.02	.18	-.03	.05	.09	.01
Que clause cont. by adjective of ease or difficulty	.00	.14	.04	.01	-.03	-.04
Que clause controlled by adjective of evaluation	.24	.15	-.06	-.05	.05	-.03
Que clause controlled by adverb	.34	-.04	.18	-.04	.09	.01
Que clause controlled by attitude adjective	.05	.05	.01	-.04	.01	.00
Que clause controlled by attitude noun	.01	.10	-.01	.01	-.17	.04
Que clause controlled by cognitive verb	.11	.28	-.06	-.05	-.01	.06
Que clause controlled by desire verb	-.11	.17	.03	-.01	-.02	.06
Que clause controlled by factive adjective	.05	.15	-.02	-.06	.02	-.01
Que clause controlled by factual noun	-.19	.26	.06	.00	.00	.03
Que clause controlled by likelihood noun	-.06	.29	.11	.03	-.04	.06
Que clause controlled by non-factual noun	-.14	.06	.07	-.03	.00	.08
Que clause controlled by noun	-.06	.59	.20	.00	.07	.31
Que clause controlled by preposition	-.27	.43	.09	.23	-.02	.38
Que clause controlled by speech verb	-.07	.17	.06	-.02	-.05	.02
Que clause controlled by verb in indicative mood	.47	.05	.32	-.20	-.15	-.01
Que clause controlled by verb in subjunctive mood	.00	.09	.11	-.03	-.03	.04
Que or infinitive clause controlled by noun	-.09	.45	.05	-.04	.01	.08
Reduced progressive clause	-.31	-.04	-.01	.18	.09	.23
Subordinating (concessive)	.15	.13	.10	-.01	.05	.17
Subordinating (conditional)	.31	.16	.23	.07	.39	.14
Subordinating (conformative)	.00	.13	.07	.02	.01	.05
Subordinating (final)	-.11	.07	-.02	.14	.07	.41
Subordinating (temporal)	-.09	.27	.16	.06	-.02	.19

(Continued)

Feature	F1	F2	F3	F4	F5	F6
Conjunctions: Coordinating (<i>ou</i>)	-.07	.04	.07	.04	.61	-.01
Conjunctions: Coordinating (adversative)	.28	.31	.07	-.04	-.19	-.02
Conjunctions: Coordinating (clausal)	.29	.12	.09	.47	-.01	.11
Conjunctions: Coordinating (conclusive)	.17	.17	.47	.05	.06	-.01
Conjunctions: Coordinating (phrasal)	-.29	.10	-.20	.04	.32	-.10
Modals: <i>Conseguir</i> (be able to)	.12	.12	.03	.15	.01	.10
Modals: <i>Dever</i> (must/should/will)	-.04	.09	.13	.03	.47	.04
Modals: <i>Haver que/haver de</i> (have to, ought to)	.03	.08	-.14	.05	-.06	.31
Modals: <i>Parecer</i> (seem to)	.21	-.03	.14	-.04	-.06	-.01
Modals: <i>Poder</i> (can)	.22	.34	.06	.05	.43	.02
Modals: <i>Precisar</i> (need to)	.34	.19	-.09	.02	.07	-.05
Modals: <i>Ter que/ter de</i> (have to, ought to)	.30	.05	.32	-.08	-.01	-.08
Nouns: Abstract	-.52	.29	.07	-.33	.29	-.19
Nouns: Animate	.05	.08	.23	-.13	-.18	.15
Nouns: Cognition	.09	.45	-.12	-.14	.21	.11
Nouns: Compound	-.65	-.05	.10	.29	.07	-.30
Nouns: Concrete	-.30	-.13	-.01	.56	-.07	.00
Nouns: Institution	-.23	.07	.26	-.18	-.17	-.09
Nouns: Nominalization in subject position	-.51	.30	.05	-.04	.14	-.12
Nouns: Place	-.17	.06	-.06	-.14	-.30	-.18
Nouns: Quantity	.05	-.15	-.14	-.11	.13	-.24
Nouns: Technical	-.29	.07	.04	.25	.06	.05
Other: Agentless passive	-.38	-.05	.11	-.11	-.11	-.08
Other: Contractions	.23	-.22	.71	.01	.16	-.24
Other: Discourse marker	.35	.00	.67	.02	.10	-.13
Other: Focus marker	.29	-.01	.28	-.05	-.07	-.12
Other: Subject omission	.34	-.07	.07	.55	-.13	.18
Prepositions: All	-.78	.01	-.15	-.04	-.03	.00
Pronouns: Demonstrative	.29	.41	.36	-.12	.10	.01
Pronouns: First person plural, in subject position	-.01	.12	.23	-.01	-.03	.05
Pronouns: First person singular, in subject position	.63	-.13	.31	-.09	.13	-.01
Pronouns: First person, object position	.62	-.10	-.14	-.07	.05	.16
Pronouns: Nominal in subject position	.31	.20	-.12	.04	-.11	.14
Pronouns: Possessive	.60	.09	-.43	.01	.08	.42

(Continued)

Table A2. (Continued)

Feature	F1	F2	F3	F4	F5	F6
Pronouns: Quantifier	.62	.35	.00	.12	-.11	-.18
Pronouns: Rare in object position	.06	.12	-.22	.17	-.02	.63
Pronouns: Relative <i>qual</i> or <i>cujo</i>	-.32	.09	-.04	-.17	.03	.06
Pronouns: Relative <i>que</i>	.01	.53	.17	-.08	.02	.34
Pronouns: Relative preceded by preposition	-.20	.30	-.05	-.14	.03	.19
Pronouns: Second person plural, in subject position	.20	.02	.05	-.03	-.03	-.01
Pronouns: Second person singular, in subject position	.76	.03	.01	.03	.29	-.09
Pronouns: Second person, object position	.91	-.06	-.13	-.03	.28	-.06
Pronouns: Third person plural, in subject position	-.03	.10	.48	.05	-.05	-.03
Pronouns: Third person singular, in subject position	.24	-.09	.50	.02	-.07	.06
Pronouns: Third person, object position	.11	.34	-.05	.05	.00	.37
Questions: <i>QU</i> questions	.54	-.14	.13	-.02	-.10	-.20
Questions: Tag questions	.02	-.16	.80	.05	.10	-.25
Questions: Yes or No question	.42	-.22	.55	-.03	-.01	-.33
Verbs: <i>Ir</i> future	.82	-.15	.06	-.13	.11	-.04
Verbs: Action	.69	.01	.09	.11	.10	.19
Verbs: Communication	.36	-.05	.15	-.22	-.16	.26
Verbs: Facilitation	-.10	.14	.07	.48	.12	.11
Verbs: First person	.86	-.06	-.01	-.13	-.03	.00
Verbs: Future present tense	.16	.07	-.23	-.11	.51	.07
Verbs: Future preterit tense	.02	.31	.03	-.13	-.22	.07
Verbs: Future subjunctive mood	.32	.09	.04	.25	.62	.17
Verbs: Gerund form, all	.34	-.07	.05	.01	.08	.17
Verbs: Imperative mood	.04	-.12	-.09	.77	.10	.18
Verbs: Imperfect	.01	.10	.26	-.04	-.38	.19
Verbs: Infinitive	.41	.12	-.05	.04	.29	.17
Verbs: Mental	.83	.10	.11	-.22	.08	.02
Verbs: Occurrence	-.02	.30	-.04	-.01	.00	.11
Verbs: Past indicative tense	.20	-.11	.09	-.21	-.55	.17
Verbs: Past participle	-.50	-.04	.03	-.15	.25	.04
Verbs: Past subjunctive mood	.05	.25	.10	-.02	-.31	.19
Verbs: Present subjunctive mood	-.08	.15	.01	.82	.26	.25
Verbs: Private	.70	.15	.22	-.17	.13	.08

(Continued)

Feature	F1	F2	F3	F4	F5	F6
Verbs: Progressive	.20	.10	.17	-.10	-.02	.05
Verbs: Progressive preceded by infinitive	.33	.00	-.10	-.10	.09	.03
Verbs: Public	.21	-.04	.09	-.25	-.17	.33
Verbs: Second person	.23	-.17	-.17	.13	.05	.47
Verbs: Split auxiliary	-.07	.10	.11	-.04	.15	-.01
Verbs: Suasive	.16	.07	-.04	-.10	.13	.21
Vocabulary distribution: Average word length	-.53	.11	-.11	-.19	.22	-.20
Vocabulary distribution: Type–token ratio	.24	-.20	-.35	.26	-.02	-.16
Vocabulary distribution: Word count	-.23	.19	.20	-.07	.01	.12

Table A3. Eigenvalues and variance accounted for by the first 10 factors

Factor	Eigenvalue	% of Variance	Cumulative %
1	22.514	15.316	15.316
2	6.554	4.458	19.774
3	5.619	3.823	23.597
4	4.257	2.896	26.493
5	3.606	2.453	28.946
6	3.039	2.067	31.014
7	3.003	2.043	33.056
8	2.8	1.905	34.961
9	2.527	1.719	36.68
10	2.197	1.495	38.175

Table A4. Interfactor correlations

Factor	1	2	3	4	5	6
1	1.000	.165	.482	.235	-.412	.156
2	.165	1.000	.045	-.027	-.119	-.148
3	.482	.045	1.000	-.112	-.362	.232
4	.235	-.027	-.112	1.000	-.056	-.193
5	-.412	-.119	-.362	-.056	1.000	-.200
6	.156	-.148	.232	-.193	-.200	1.000